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# Security Guide for SAP Service Cloud Version 2

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# 1 **Security Guide for SAP Service Cloud Version 2**

This guide provides an overview of the security-relevant information that applies to SAP Service Cloud Version 2.

## **Why Is Security Necessary?**

With the increasing use of distributed systems and the Internet for managing business data, the demands on security are also on the rise. When using a distributed system, you need to be sure that your data and processes support your business needs without allowing unauthorized access to critical information. User errors, negligence, or attempted manipulation of your system should not result in loss of information or processing time. While it is primarily the customer's responsibility to ensure their data security and proper user management, SAP supports security by providing relevant features and functions. SAP is also responsible for managing the lifecycle of the application for security improvement.

To assist you in securing SAP Service Cloud Version 2, we provide this Security Guide.

## 2 Technical System Landscape

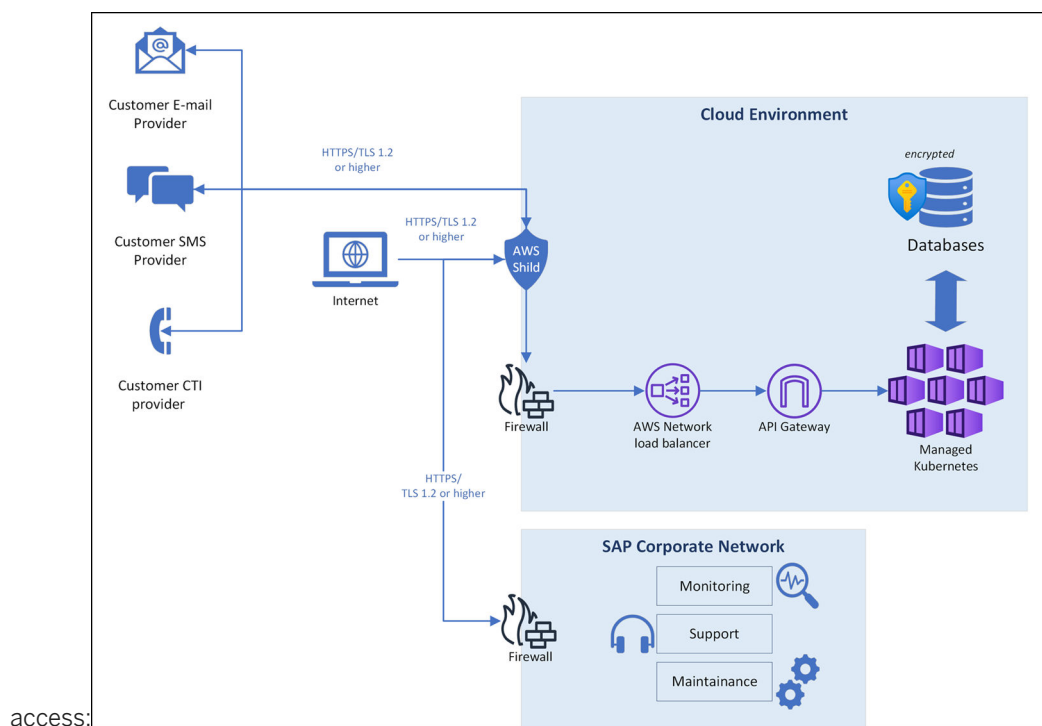
SAP data centers provide high level security measures while still allowing integration and flexible access to their cloud data.

SAP Cloud native solutions are hosted in Amazon Data Centers and operated by SAP. You, as a customer, can select the data center region that best fit your needs.

Since SAP Cloud solutions deal with business data from your core business processes, SAP adheres to the highest security and quality requirements, as follows:

- All communication with the solution flows through secure channels (HTTPS/TLS1.2 as the minimum).
- SAP-controlled keys secure all rest data using AES 256 encryption.
- As stated in the Data Processing Agreements (DPA), all data belongs to you - SAP processes the data on your behalf.
- When you share your infrastructure, your data is separated into tenants. Technical measures ensure that cross-tenant access isn't possible.
- Infrastructure events are collected and sent to a centrally monitored SIEM.
- Infrastructure is regularly scanned to detect policy violations and harden existing security guidelines.
- To access the customer tenants, you must enter a unique, customer-specific URL (UI Client or APIs).
- Users are unique and authentication (username/password, certificate based or single-sign-on) is mandatory.
- Role-based access management is provided to support granular controls, segregation of duties, and administration.

The following diagram summarizes the technical system landscape for standard



## 3 Users and Control

Users and control settings allow you, as an administrator, to manage your employees, users, business roles, and security policies which in turn provide authentication and authorization services for your system.

- **Employees:** A group of people who contribute to creating goods or services in a company, based on a work contract or a contract for services. Unlike externals, internal employees are bound by instructions and obliged to adhere to the company's policies and regulations.
- **Users:** Refer to employees, business users, and other users such as partner contacts. Users are assigned to business roles and technically, access controls in your system are implemented on the business roles.
- **Business Roles:** Represents a set of authorizations to access your solution within your organization and system landscape. In the system, a business role defines a hierarchy of access controls that you can assign to application users in your organization. You can use roles to assign access to a large number of users at a time. As roles change, you need to update only the access rights within a role and the rights apply for all assigned users.
- **Security Policies:** A set of rules that defines password complexity, such as including numerical digits and password validity, like requiring a password change after a certain period of time.

### Authentication and Authorization

Authentication is the process of verifying who a user is, while authorization is the process of verifying what a user will have access to.

The different authentication methods include:

- **Basic Authentication:** Basic authentication is a simple authentication method that is built into the HTTP protocol that is supported by most Web browsers and Web servers. With this authentication mechanism, the client sends the user ID and password to the server in plain text format.
- **Single Sign-On Authentication:** Single sign-on (SSO) is an authentication method that enables users to securely authenticate with multiple applications and websites by using just one set of credentials. It is a mechanism that eliminates the need for users to enter passwords for every system that they log on to.
- **Certificate-Based Authentication:** Certificate-based authentication is a cryptographic technique that enables secure identification of one computer by another across a network connection. It uses a public-key certificate. This authentication system confirms a user's or device's identity using digital certificates issued by a trusted authority such as a government agency or web server to verify its authenticity.
- **Token-Based Authentication:** Token-based authentication is a protocol which allows users to verify their identity, and in return receive a unique access token. Until the token is valid, the users can access the website or app that the token has been issued for without having to re-enter credentials each time they go back to the same webpage or app. The user retains access as long as the token remains valid. Once the user logs out or quits an app, the token becomes invalid.

For example: If you want to do single call where you intend to read case and opportunity, you call the opportunity API by passing basic auth headers (User Name and Password). If you want to make like 100s of calls, then it is best to use Auth Token. You use Auth Token call and provide details for Basic Authentication, with which you get an Access Token. Once you have the Access Token, you don't need basic authentication credential each time you want to access any APIs, instead you use bearer token. In

the bearer token, you can pass the Access Token and its validity. Until the validity expires, the authorization remains valid.

## 3.1 Users

Users are created for system access. Business users are created for employees to access the system through UI, whereas API users are technical users who can access the system through APIs.

In this section, we discuss about the different types of users that exist in the solution.

### User Types

The different types of users in the system include:

- **Business User:** Users who have at least one business role assigned to them that governs the access to the system. These users can access the user interface of the application.
- **Technical User:** Users that have been created for a specific purpose. The different types of technical users include:
  - **Communication User** – Users created and used for integration purposes only. A communication user is used only for accessing a specific communication scenario for which they've been created. They have API access, but not UI access.
  - **Support User** – Users who are used by support teams to log on to customer tenants to access the system as part of case processing.

#### Note

Support teams may receive a case that requires access to the customer system. This scenario can occur if the issue can't be reproduced or resolved in internal environments. In such situations, the Cloud Access Manager (CAM) is used to generate temporary access to the corresponding customer system.

Support users aren't allowed to share these details. The CAM tool maintains a log of which user generated which support user at what date and time. So, it's always possible to link a generic support user back to the real person.

Support users follow the pattern **SAP\_\***.

- **Initial User** – Users who are created initially when a tenant is provisioned, and are later locked.
- **API User** – Users created and used for consuming other services through APIs. These users have API access as per the technical roles assigned. They don't have access to the UI of the application.

#### Note

For communication and API users, only for the first logon for the day, an event is logged in and published in the log.

- **Active Users:** It's a metric, and is defined as the number of unique users who are logging in a 30-day period.

- **Licensed Users:** It's a metric, and is defined as the total number of valid Business Users in the system.
- **Service Account User:** These users are created at a tenant Level. Displayed in the system as `SAP_SYSTEM01` is a service account user. They can't be created using HTTP APIs but are auto-created by IAM services. Some of the features of such service account users include:
  - `SAP_SYSTEM01` user is created in response to a tenant create event.
  - `SAP_SYSTEM01` user can't be used to log in to the tenant.
  - `SAP_SYSTEM01` user is used for any non-interactive user access and updates in the tenant that ensures that all access and updates are always based on a valid tenant user.
  - Each tenant contains a Service Account User (`SAP_SYSTEM01`) that is created with an all-zero UUID (00000000-0000-0000-0000-000000000000).
- **Limited User:** A user who can perform a limited number of transactions in the tenant. The total number of transactions that a limited user can perform are 1200 transactions per calendar year. To create a limited user, an administrator must choose the [Limited User](#) SKU in the [Manage SKUs](#) section while creating a business role. Once the limited user SKU is selected, you can't add any other SKU. Any user assigned to this role is counted as a limited user.  
The transactions that get reported when a limited user can perform actions include:
  - Cases
  - Phone Calls
  - Emails
  - Tasks
  - Chats
  - Notes
  - Leads
  - Opportunities
  - Appointments
  - Visits
  - Attachments
  - Quotes
  - Orders

## Business User List View

You can view all the business users in your system in a list format.

The following header fields are available on this page:

- **User Name:** the user name of your business user.
- **Full Name:** the full name of your business user.
- **Created On:** the calendar date on which your business user was created.
- **Valid Until:** the calendar date until which your business user is valid in the system.

### Note

The user validity is independent of Employee validity. It means an employee valid until date isn't the same as the user valid until date.



## Technical User List View

You can view all the technical users in your system in a list format.

The following header fields are available on this page:

- User Name: the user name of your technical user.
- User Type: the type of technical user.
- Valid From: the calendar date on which your technical user was created.
- Valid Until: the calendar date until which your technical user is valid in the system.

### Note

User validity is independent of Employee validity. It means an employee valid until date isn't the same as the user valid until date.

- Security Policy: the security policy associated with the user.

## 3.1.1 Business User Details

View details of a business user on this page.

The *Details* tab provides the following information of a business user.

- *Employee Data*: Shows the basic employee information such as Employee Name, Employee ID, Company, Email ID, and so on.
- *User Details*: Displays the following user information.
  - *User Name*: A unique name to identify a business user. Only an administrator has the privileges to edit it.
  - *Global User ID*: A unique identifier for a user for content records.
  - *Created On*: The date on which the business user was created in the system.
  - *Valid From*: The date from which the business user is valid. An administrator can edit this date.

### Note

*Valid From* date may not be the same as *Created On* date. *Valid From* date can be a future date. A business user is a valid user only between the interval defined by *Valid From* and *Valid Until* dates. An administrator can edit the *Valid From* date.

- *Valid Until*: The date until which the user is active in the system.
- *Password Locked*: A switch that indicates if the password is locked or not. The following are the features of the *Password Locked* switch:
  - The switch is auto-enabled after too many failed login attempts.
  - If the switch is enabled, an employee can't log on to the system from the login screen but can only log in using a certificate.
  - An administrator can disable this switch to unlock the password, however, an administrator can't lock the password.
- *Security Policy*: Displays the security policy assigned to the user. An administrator can change the assigned security policy.

- **Work Time:** Edit the time zone in which your business users work. An administrator can set the work time for business users.

#### Note

The **Work Time** field is for Dynamic Visit Planning functionality purposes and is used for visit planning and scheduling process. This settings won't impact the regular time zone configuration of the system.

- **SCIM Replication:** A flag that indicates if the user is locally created or SCIM replicated.
- **Reset Password:** Provides an option to reset the current password of the user. As an administrator, you can reset a user's password by clicking the **Reset Password** button, which triggers a password reset email to your user. The user can then reset the password using the **Password Reset** link in the email.

#### Note

- The **Password Reset** link is valid for 60 minutes and is for single password reset only.
- You can view user created and password reset email details in **Outbound Monitoring** setting.

- **Manage Certificates:** Enables an administrator to add or remove certificates for a user to enable certificate-based login for a user.

To add a new certificate for a user:

1. Choose **Manage Certificates**.
2. Upload a certificate for a user using the **Drop or Browse Files** option. Once the certificate is uploaded, it enables an user with seamless login to the system.

#### Note

You can upload only certificates generated from the certificate authority listed in the following note: <https://launchpad.support.sap.com/#/notes/2801396>.

To remove a certificate, select the certificate and select the delete icon under the **Actions** column.

The **Assigned Business Roles** tab lists all the business roles assigned to a particular business user. An administrator can search and add a business role. The administrator can also delete an assigned role.

## Related Information

### [Users \[page 7\]](#)

Users are created for system access. Business users are created for employees to access the system through UI, whereas API users are technical users who can access the system through APIs.

### [Create Business Users and Assign Business Roles \[page 11\]](#)

Administrators can create business users and provide appropriate authorizations by assigning business roles.

## 3.1.2 Create Business Users and Assign Business Roles

Administrators can create business users and provide appropriate authorizations by assigning business roles.

### Context

Business users in the system are linked to employees. Business users have business roles assigned, and the access controls in the system are implemented on business roles.

### Procedure

1. Log in to the system as an administrator.
2. Navigate to [User Menu](#) on the top-right corner, and access the [System Settings](#) page.
3. Go to ► [All Settings](#) ► [Users and Control](#) ► [Users](#) ►.
4. Choose [Business Users](#) from the dropdown on the page.
5. Click the Add icon ( + ) on the [Business Users](#) page.
6. Search and select an employee for whom you wish to create the business user.

#### Note

- The employee linked to a user must have a unique Email ID.
- If a business user already exists for the employee, a message stating *The business user already exists for this employee* is displayed.

7. Once you select an employee, the [User Data](#) section is auto-populated. By default, the user name is created with the last name followed by the first name of the employee. If you wish to modify the user name, you can modify them.
8. Once you've verified all the details, click [Create](#). The business user is now created and linked to the employee. Refresh the business users' list to view your newly created business user.  
  
Once the user is created, an email is triggered to the users. This email contains the username, a link to set the password, and the system URL. For SSO-enabled users, an SSO-enabled URL is sent in the email, using which the users can log in via SSO to the system.
9. Choose your business user, and then navigate to the [Assigned Business Roles](#) tab.
10. Add business roles to your user by using the [Search and Add](#) functionality on the screen. All the business roles that you assigned to your business user now appear in the list format on the [Assigned Business Roles](#) tab.

#### Note

- Ensure that your administrator has access to the role service in addition to the user service to assign roles to your business users.
- If a role assigned to your user has its [Write Access](#) set to [No Access](#) in the [Business Services](#) section, then such a user can't edit business roles.

You've now created a business user and assigned the user to a business role.

## Related Information

[Business User Details \[page 9\]](#)

View details of a business user on this page.

### 3.1.3 Technical User Details

View details of a technical users on this page.

The [Details](#) tab provides the following information of a technical user:

- [User Details](#): Displays the following user information.
  - [User Name](#): A unique name to identify a technical user. Only an administrator has the privileges to edit it.
  - [Display Name](#): A unique identifier for a user for content records.
  - [Created On](#): The date on which the technical user was created in the system.
  - [Valid Until](#): The date until which the user is active in the system. An administrator can edit this date.
  - [Password Locked](#): A switch that indicates if the password is locked or not. The following are the features of the [Password Locked](#) switch.
    - The switch is auto-enabled after too many failed login attempts.
    - An administrator can disable this switch to unlock the password, however, an administrator can't lock the password.
- [Security Policy](#): Displays the security policy assigned to the user. An administrator can change the assigned security policy.
- [Change Password](#): Provides an option to change the current password of the user. As an administrator, you can change a user's password by clicking the [Change Password](#) button.
- [Manage Certificates](#): Enables an administrator to add or remove certificates for a user to enable certificate-based login for a technical user.

To add a new certificate for a technical user:

1. Choose [Manage Certificates](#).
2. Upload a certificate for a user using the [Drop or Browse Files](#) option. Once the certificate is uploaded, it enables an user with seamless login to the system.

#### Note

You can upload only certificates generated from the certificate authority listed in the following note:  
<https://launchpad.support.sap.com/#/notes/2801396> .

3. To remove a certificate, select the certificate and click the delete icon under the [Actions](#) column.

The [Assigned Technical Roles](#) tab lists all the technical roles assigned to a particular API user. An administrator can search and add a technical role. The administrator can also delete an assigned role.

## Related Information

[Users \[page 7\]](#)

Users are created for system access. Business users are created for employees to access the system through UI, whereas API users are technical users who can access the system through APIs.

[Create Technical Users and Assign Technical Roles \[page 13\]](#)

Administrators can create API users and provide authorizations by assigning them to technical roles.

## 3.1.4 Create Technical Users and Assign Technical Roles

Administrators can create API users and provide authorizations by assigning them to technical roles.

### Context

An API user is a type of technical user who is authorized to consume other services through APIs. These users can access the system using the APIs as per the technical roles assigned to them. They don't have access to the UI of the application. For an API user, an Employee can't be assigned, and all the details of the user must be entered manually.

### Procedure

1. Log in to the system as an administrator.
2. Navigate to the [User Menu](#) on the top-right corner, and access the [System Settings](#) page.
3. Go to ► [All Settings](#) ► [Users and Control](#) ► [Users](#) ►.
4. Choose [Technical Users](#) from the dropdown on the page.
5. Click the Add icon (+) on the [All Technical Users](#) page, and enter a [User Name](#), [Display Name](#), [Password](#), [Valid From](#), and browse to upload a certificate on the [Create API User](#) window. By default, a security policy relevant for the technical user is assigned.
6. Click [Create](#). The API user is now created. Refresh the [All Technical Users](#) page to view your newly created API user.
7. Choose your technical user, and then navigate to the [Assigned Technical Roles](#) tab.
8. Add technical roles to your user by using the [Search and Add Technical Roles](#) button on the screen. All the technical roles that you assigned to your API user now appear in a list format on the [Assigned Technical Roles](#) tab.

#### Note

- Ensure that your administrator has access to the role service in addition to the user service to assign roles to your technical users.
- If a role assigned to your user has its [Write Access](#) set to [No Access](#) in the [Business Services](#) section, then such a user can't edit business roles.

You've now created an API user and assigned a technical role to the user. You can now use this user to consume other services through APIs.

## 3.2 Roles

A business role defines the specific areas of the system the users can access when they're assigned to that role. Administrators can implement authorizations to business users by assigning business roles.

A business role represents a set of authorizations to access your solution within your organization and system landscape. A business role defines a hierarchy of access controls that you can assign to business users in your organization. As roles change, you need to update only the access rights within a role and the rights will apply for all assigned users.

### Business Roles List View

Lists all the business roles in a list view. As an administrator, you can edit the following fields on this page:

- Business Role ID
- Business Role Name
- Description
- Status: You can set the status to *In Preparation*, *Active*, or *Obsolete*.
- Admin: You can activate this switch to indicate that the role is an administrator role.

### Technical Roles List View

Lists all the technical roles in a list view. As an administrator, you can edit the following fields on this page:

- Technical Role ID
- Technical Role Name
- Description
- Status: You can set the status to *In Preparation*, *Active*, or *Obsolete*.
- Admin: You can activate this switch to indicate that the role is an administrator role.

### 3.2.1 Business Role Details

View more details of a business role on this page.

The *Details* tab provides the following information of a business role.

- **Business Services:**

- All business services that are assigned to a business role are displayed in the [Business Services](#) list. As an administrator, you can add or delete business services to the business role.
- All business services have unrestricted read and write access by default.
- Some business services allow creation of restrictions. Such services have [Read/Write](#) dropdown list enabled. On setting any one of them to [Restricted](#), the [Restriction Rule](#) link is enabled.

#### Note

If one role blocks write access and another role allows it, the more permissive role wins.

- An administrator can select from the available restrictions for a given service to create a restriction rule.
- You can delete invalid business services that get flagged with the following message "Some of the business services assigned to the business role are no longer valid. Please delete all the services with a warning sign to proceed further."
- Default and Mandatory Business Services: When a business role is created, a few of the services such as [Org. Units](#), [Data Import and Export](#), [Home Page](#), [Service Catalog](#), and so on, are assigned by default. These services are default (non-mandatory) business services. The respective apps are added to the app list. These services can be removed by an administrator while either creating or editing a business role. Also, some of the business services such as [Search](#) and [Language, Region, and Currency](#) services are also assigned by default. These services and their respective apps can't be removed or disabled by an administrator. These services are mandatory business services for a role.
- **App List:**
  - Each business service can have one or more apps associated with it. These apps define the left navigation for the users having this particular business role assigned.
  - On adding the business service to a role, the corresponding apps are added to the app list.
  - Administrators can enable or disable these apps using the switch against them.
  - You can also search for apps from the search bar. You can also filter the app list by [All Apps](#), [Generic Apps](#), and [Admin Apps](#).
  - As an administrator, you can also enable all apps at once using the [Enable All Apps](#) switch and enable all Admin Apps at once using the [Enable All Admin Apps](#) switch.

#### Note

The overall access to entities is controlled and governed by the [Business Services](#) section. However, access to a particular entity on the UI is managed using [Apps](#). If you want to provide restricted access to a particular entity, use the restriction rules in the [Business Services](#) section.

The [Assigned Business Users](#) tab lists all the users that have this business role assigned. An administrator can search and add users to this role. The administrator can also delete users from this role.

The [Changes](#) tab lists all the changes made to your business role. These changes include the modifications made to restriction rules for a particular service. The changes display the business service name that is affected when a restriction rule is edited. You can use the different filters available on the interface to filter and view the changes.

## Related Information

[Changes \[page 60\]](#)

Review detailed configuration changes with effective filtering capability.

### 3.2.2 Create Business Roles and Assign Business Users

Administrators can create business roles with appropriate authorizations and assign them to business users.

#### Context

Use business roles to assign access rights to multiple business users who carry out similar activities. You can also define access restrictions for a business role and in turn to the business users that are assigned to the role.

#### Procedure

1. Navigate to [User Menu](#) on the top-right corner, and access the [System Settings](#) page.
2. Go to ► [All Settings](#) ► [Users and Control](#) ► [Roles](#) ►.
3. Choose [Business Roles](#) from the dropdown list.
4. Click the Add (+) icon on the [All Business Roles](#) page, and enter a business role name and description. The [Business Role ID](#) is auto-generated. You can also set this role as an admin role by enabling the [Admin](#) switch.
5. Click the Add (+) icon in the [Manage SKUs](#) section, and choose one or more SKUs from the [Select SKUs](#) list. However, if you choose Limited User SKU from the list, you can't select or add any other SKUs from the list. Also, you can't add Limited User SKU if any other SKU is already added.

#### Note

- In general, you can select more than one SKU from the list. You can remove SKUs that you no longer need. In such a case, the corresponding services are removed from the role. The business service is retained only if it's also part of another assigned SKU.
- If you select the limited use SKU from the [Select SKUs](#) list, a list of all business services (associated with all other existing SKUs in the tenant) is fetched and enabled, except for the Groupware services.

6. Based on the selected SKUs, a list of services is populated by default. The list of business services is filtered based on the selection of SKUs. Click the Add (+) icon in the [Business Services](#) section to open the [Add Business Services](#) popup, select all the services that are required for this role from the list of services, and then click [Add](#). You can now [Save and Activate](#) your business role.



### Note

- Choosing an SKU in the [Manage SKUs](#) section is applicable for new roles only and doesn't impact the existing business roles.
- **Mandatory and Non-Mandatory Services:**  
Some business services such as [Search](#) and [Language, Region, and Currency](#) business services are also assigned by default. These services and their respective apps can't either be removed or disabled by an administrator. These services are mandatory business services for a business role. If you try to remove these services or disable these apps while creating or editing a business role, they're added back again to the role. For an existing business role, this change has no functional impact.  
By default, a list of business services such as [Org. Units](#), [Data Import and Export](#), [Home Page](#), [Service Catalog](#), and so on, is assigned. Such services are default (or non mandatory) business services. The respective apps are also added to the app list. These services can be removed by an administrator either while creating or editing a business role.
- If you set a role as an Admin role using the switch, a default list of business services necessary for the administrator role is added by default.

7. Click the Add (+) icon in the [Business Services](#) section, and select and add all the services that are required for this role from the [Add Business Services](#) popup. [Save and Activate](#) your business role.

### Note

If you're editing a business role, you must explicitly use the [Save](#) option to ensure all your changes get saved on both [Business Role Details](#) and [Assigned Business Users](#) pages.

8. You can add additional services for your role from the [Add Business Services](#) popup.
9. Set restrictions for the business services by editing either or both [Read Access](#) and [Write Access](#) columns for a corresponding business service. By default, both these columns are set to unrestricted for a service, and the [Restriction Rule](#) is set to unassigned. Once you set [Read Access](#) and [Write Access](#) for a business service, the corresponding [Restriction Rule](#) is enabled. Click the enabled [Unassigned](#) button to set access restriction rules for your business role, based on the options listed.

### Note

**Access Restriction Rules:** Based on the restriction types defined by each service, the following restriction rules are supported. When a business role is maintained for a user, based on supported restriction types for a service, the corresponding rules are displayed. You can select one or more rules.

#### Access Restriction Rules

| Restriction Type | Restriction Rule          | Description                                                                                                                                                         |
|------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EMPLOYEE         | Myself                    | This rule provides access based on direct employee assignment                                                                                                       |
|                  | Employees Reporting to Me | For a manager: Access is based on employee and involvement of employees reporting to a user in an org. unit (including subunits). For a non-manager: Same as Myself |
|                  | Specific Employees        | This rule provides access to specific employees                                                                                                                     |

| Restriction Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Restriction Rule              | Description                                                                                             |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|---------------------------------------------------------------------------------------------------------|
| ORGANIZATIONAL UNIT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | My Service Organization       | This rule provides access to the service organization of the employee in the organizational hierarchy   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | My Sales Organization         | This rule provides access to the sales organization of the employee in the organizational hierarchy     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | My Organizational Unit        | This rule provides access to an organizational unit of the employee in the organizational hierarchy     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Specific Organizational Unit  | This rule provides access to specific organizational units (which the employee doesn't belong to)       |
| <div>  <b>Note</b><br/> As a prerequisite, the organizational unit must be set as the <i>Trusted Authorization Unit</i> in the account or individual customer details. The <i>Trusted Authorization Unit</i> field is enabled via adaptation. Restricting a business role for an account or individual customer service by the organizational unit only affects the account or individual where the <i>Trusted Authorization Unit</i> field is maintained. If the <i>Trusted Authorization Unit</i> is specified for an account or individual customer, the account can't be restricted by the employee, territory, or sales data. </div> |                               |                                                                                                         |
| TERRITORY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | My Territory Hierarchy        | This rule provides access based on the employee's territory assignment only (including sub-territories) |
| SALES AREA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | My Sales Data                 | This rule provides access based on employee's sales data (sales org. + division + distribution channel) |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Sales Areas of My Sales Orgs. | This rule provides access based on all sales areas of sales organizations of employee                   |

| Restriction Type   | Restriction Rule                      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DISTRIBUTION CHAIN | My Sales Data                         | <p>This rule refers to a distribution chain maintained at an employee level. Provides access based on the distribution chain.</p> <p>A distribution chain is a combination of Sales Org. and Distribution Channel code</p> <ul style="list-style-type: none"> <li>• A sales organization (for example, Sales Organization France)</li> <li>• A distribution channel code (for example, 1 = direct sales and 2 = indirect sales)</li> </ul> <p>Therefore, the distribution chain at the employee level has the sales data of:</p> <ul style="list-style-type: none"> <li>• Sales Organization France 1</li> <li>• Sales Organization France 2</li> </ul> |
|                    | Distribution Chains of My Sales Orgs. | <p>If an employee is assigned under Sales Organization France and once you select this restriction, you get access for any distribution chains with Sales Organization France that is Sales Organization France *Refers to the Org. with any distribution channel. The restriction rule checks to which Sales Org. the user is assigned. Accordingly, any distribution chain with this Sales Org. is allowed (combination of the user's Sales Org. with any Distribution Channel)</p>                                                                                                                                                                   |

#### Note

If one rule blocks the Write Access and another role allows it, the more permissive role takes precedence.

- The [App List](#) section, following the [Business Services](#) section, displays all the available apps for the selected business services. Select the switch against an app to make it available for the business role.

#### Note

The overall access to entities is controlled and governed by the [Business Services](#) section. However, access to a particular entity on the UI is managed using [Apps](#). If you want to provide restricted access to a particular entity, use the restriction rules in the [Business Services](#) section.

- Navigate to the [Assigned Users](#) tab. A list of users who have been this role assigned is displayed. If there are no business users assigned yet, the [Search and Add Users](#) button is visible.

- Click the [Search and Add Users](#) button to add business users from the quick view. Search and select users from the quick view to add them to the role.

#### ⓘ Note

- Ensure that your administrator has access to the user service in addition to the role service to assign users to your business roles.
- You can't assign business users if [Write Access](#) is set to [No Access](#) for [User](#) service. If otherwise, the user can't edit the role but can edit the user assignments.

## Results

You've now created a business role and assigned the business role to the selected list of business users.

## Related Information

[Business Role Details \[page 14\]](#)

View more details of a business role on this page.

[List of Business Services \[page 22\]](#)

This document provides a list of business services available in SAP Sales Cloud Version 2 and SAP Service Cloud Version 2.

[Users \[page 7\]](#)

Users are created for system access. Business users are created for employees to access the system through UI, whereas API users are technical users who can access the system through APIs.

## 3.2.3 Copy Business Roles

Create copy of business roles and assign it to your business users.

## Context

Administrators can now create copy of business roles using the inline copy function on the object worklist page.

## Procedure

- On the [All Business Roles](#) page, choose the inline [Copy](#) option for a business role that you wish to copy. A new quick create page, [Copy Role](#) opens up.

#### Note

A new role ID is generated, and the role name is populated as *Copy Of <the existing business role>*, which can be edited to give a name.

2. Enter a description for the role and choose *Save and Open* to open the new business role. All the business services and the apps assigned to the role are copied as well.

#### Note

The new business role is set to active status by default, and if your role had administrator authorizations, the same is retained. However, none of the users assigned to the role is retained. Ensure that you assign users to your business role.

## Results

Your role is now copied and ready. You can assign business users to your role.

## 3.2.4 Create Technical Roles and Assign Technical Users

Administrators can create technical roles with appropriate authorizations and assign them to technical API users to consume services.

## Context

Use technical roles to assign access rights to multiple technical API users who carry out similar activities.

## Procedure

1. Log in to the system as an administrator.
2. Navigate to *User Menu* on the top-right corner, and access the *System Settings* page.
3. Go to ► *All Settings* ► *Users and Control* ► *Roles* ▾.
4. Choose *Technical Roles* from the dropdown on the page.
5. Click the Add icon ( + ) on the *All Technical Roles* page, and enter a technical role name and description. The *Technical Role ID* is auto-generated. You can also set this role as an admin role by enabling the *Admin* switch.
6. Click the Add icon ( + ) in the *Business Services* section to open the *Add Business Services* popup, select all the services that are required for this role from the list of services, and then click *Add*. You can now *Save and Activate* your business role.

7. Set restrictions for the newly added business services by editing either or both of the [Read Access](#) and [Write Access](#) columns for a corresponding business service. By default, both these columns are set to unrestricted for a service. You can choose to impose restrictions on these services by choosing [Unrestricted](#) for [Read Access](#) and either [Unrestricted](#) or [No Access](#) for [Write Access](#).
8. Navigate to the [Assigned Technical Users](#) tab. A list of users who have been this role assigned is displayed. If there are no technical API users assigned yet, the [Search and Add Technical Users](#) button is visible.
9. Click the [Search and Add Technical Users](#) button to add technical API users from the quick view. Search and select users from the quick view to add them to the role.

#### Note

- Ensure that your administrator has access to the user service in addition to the role service to assign users to your technical roles.
- You can't assign technical users if [Write Access](#) is set to [No Access](#) for [User](#) service. If otherwise, the user can't edit the role but can edit the user assignments.

## Results

You've now created a technical role and assigned the technical role to the selected list of technical API users.

## Related Information

### [Roles \[page 14\]](#)

A business role defines the specific areas of the system the users can access when they're assigned to that role. Administrators can implement authorizations to business users by assigning business roles.

## 3.2.5 List of Business Services

This document provides a list of business services available in SAP Sales Cloud Version 2 and SAP Service Cloud Version 2.

| Business Service Name | Description                                                                              | Business Service: Technical Details                                                                                                                                                                                      |
|-----------------------|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Accounts              | Monitor, store, and track business information about customers, prospects, and partners. | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/><code>sap.crm.md.service.accountService</code></li> <li>• <b>UI App Name:</b><br/><code>sap.crm.md.accountservice.uiapp.accountApp</code></li> </ul> |

| Business Service Name   | Description                                                                                                    | Business Service: Technical Details                                                                                                                                                                                                                                   |
|-------------------------|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Account Hierarchy       | Map complex organizational structures of a business partner or a large customer with multiple subsidiaries.    | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/><code>sap.crm.md.service.accountHierarchyService</code></li> <li>• <b>UI App Name:</b><br/><code>sap.crm.md.accounthierarchyService.uiapp.accountHierarchyApp</code></li> </ul>                   |
| Account Hierarchy Admin | Configure account hierarchy settings.                                                                          | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/><code>sap.crm.md.service.accountHierarchyService</code></li> <li>• <b>UI App Name:</b><br/><code>sap.crm.md.accounthierarchyService.uiapp.accountHierarchyAdminApp</code></li> </ul>              |
| Address                 | Add, modify, or delete addresses.                                                                              | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/><code>sap.crm.md.service.addressService</code></li> <li>• <b>UI App Name:</b><br/><code>sap.crm.md.addressservice.uiapp.addressAdminApp</code></li> </ul>                                         |
| Agent Desktop           | Review customer information, create and process cases, manage interactions, and review knowledge base articles | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/><code>sap.crm.service.agentDeskService</code></li> <li>• <b>UI App Name:</b><br/><code>sap.crm.agentdeskservice.uiapp.agentDesk</code></li> </ul>                                                 |
| Agent Desktop Settings  | Configure agent desktop settings.                                                                              | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/><code>sap.crm.service.agentDeskService</code></li> <li>• <b>UI App Name:</b><br/><code>sap.crm.agentdeskservice.uiapp.agentDeskAdmin</code></li> </ul>                                            |
| Alerts Service          | Configure alert services to receive notifications.                                                             | <b>Technical Name:</b><br><code>sap.crm.service.alertsService</code>                                                                                                                                                                                                  |
| Analytics               | Create analytical stories and models.                                                                          | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/><code>sap.crm.service.analyticsBusinessUserStoriesService</code></li> <li>• <b>UI App Name:</b><br/><code>sap.crm.analyticsbusinessuserstorieservice.uiapp.anaStoryBusinessUser</code></li> </ul> |

| Business Service Name          | Description                                                                                      | Business Service: Technical Details                                                                                                                                                                                  |
|--------------------------------|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Analytics - SAC Setup Service  | Configure SAP Analytics Cloud setup.                                                             | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.analyticsSacSPSetuService</li> <li>• <b>UI App Name:</b><br/>sap.crm.analyticsacspsetupservice.uiapp.idPSetupUI</li> </ul>       |
| Analytics - User Story Service | View stories assigned to your user.                                                              | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.analyticsUserStoriesService</li> <li>• <b>UI App Name:</b><br/>sap.crm.analyticsuserstorieservice.uiapp.anaStoryAdmin</li> </ul> |
| Application Log                | View a summary of all errors that occurred during the processing of custom logic external hooks. | <b>Technical Name:</b><br>sap.crm.md.service.accountService                                                                                                                                                          |
| Approvals                      | Create approvals for streamlining your business processes.                                       | <b>Technical Name:</b><br>sap.crm.md.service.accountService                                                                                                                                                          |
| Attachments                    | Choose the files that you want to upload or download.                                            | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.documentService</li> <li>• <b>UI App Name:</b><br/>sap.crm.documentservice.uiapp.document</li> </ul>                             |
| Attachments Admin              | Configure your attachment settings.                                                              | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b></li> <li>• <b>UI App Name:</b><br/>sap.crm.documentservice.uiapp.documentAdmin</li> </ul>                                                            |
| Audit Log                      | Review security-relevant chronological records of user actions.                                  | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b>sap.crm.service.auditService</li> <li>• <b>UI App Name:</b><br/>sap.crm.documentservice.uiapp.documentAdmin</li> </ul>                                |
| Authorization Service          | Configure authorization services.                                                                | <b>Technical Name:</b> sap.crm.service.authService                                                                                                                                                                   |
| Autoflow                       | Automate your business processes by creating rules that can trigger various events.              | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.workflowService</li> <li>• <b>UI App Name:</b><br/>sap.crm.workflowservice.uiapp.workflowAdmin</li> </ul>                        |



| Business Service Name          | Description                                                                                                  | Business Service: Technical Details                                                                                                                                                                                                                             |
|--------------------------------|--------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Built-In Support Service       | Find help, create tickets, and contact support right in your cloud application.                              | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.builtinSupportService</li> <li>• <b>UI App Name:</b><br/>sap.crm.builtinsupportservice.uiapp.builtinSupportUi</li> </ul>                                                    |
| Built-In Support Service Admin | Configure support services to find help, create tickets, and contact support right in your cloud application | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.builtinSupportService</li> <li>• <b>UI App Name:</b><br/>sap.crm.builtinsupportservice.uiapp.builtinSupportAdminUi</li> </ul>                                               |
| Business Partners              | Capture, monitor, store, and track all critical information about business partners.                         | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.md.service.businessPartnerService</li> <li>• <b>UI App Name:</b><br/>sap.crm.md.businesspartnerservice.uiapp.businessPartnerApp</li> </ul>                                          |
| Business Partners Admin        | Configure business partner settings.                                                                         | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.md.service.businessPartnerService</li> <li>• <b>UI App Name:</b><br/>sap.crm.md.businesspartnerservice.uiapp.businessPartnerAdminApp</li> </ul>                                     |
| Business Partner Relationship  | Add relationships between different business partners.                                                       | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.md.service.businessPartnerRelationshipService</li> <li>• <b>UI App Name:</b><br/>sap.crm.md.businesspartnerrelationshipservice.uiapp.businessPartnerRelationshipAdminApp</li> </ul> |
| Call Lists Administration      | Configure settings for Call lists.                                                                           | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.callListService</li> <li>• <b>UI App Name:</b><br/>sap.crm.callListService.uiapp.callListAdminApp</li> </ul>                                                                |

| Business Service Name | Description                                                                                        | Business Service: Technical Details                                                                                                                                                                                |
|-----------------------|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Call Lists            | Create call lists from the database of contacts, prospects, and leads. Also track and manage them. | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.callListService</li> <li>• <b>UI App Name:</b><br/>sap.crm.calllistservice.uiapp.callList</li> </ul>                           |
| Case                  | Create cases and manage case phases and steps.                                                     | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.caseService</li> <li>• <b>UI App Name:</b><br/>sap.crm.caseservice.uiapp.case</li> </ul>                                       |
| Case Management       | Create case types and configure case settings.                                                     | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.caseService</li> <li>• <b>UI App Name:</b><br/>sap.crm.caseservice.uiapp.caseAdmin</li> </ul>                                  |
| Case Type             | Create and configure case types to match the workflow used by your organization.                   | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.caseTypeService</li> <li>• <b>UI App Name:</b><br/>sap.crm.caseservice.uiapp.caseAdmin</li> </ul>                              |
| Change History        | Check when and by whom an entity was modified.                                                     | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.changeHistoryService</li> <li>• <b>UI App Name:</b><br/>sap.crm.service.changehistoryservice.uiapp.changeHistoryApp</li> </ul> |
| Change Project        | Record and transport configuration changes from a source tenant to a target tenant.                | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.changeProjectService</li> <li>• <b>UI App Name:</b><br/>sap.crm.changeprojectservice.uiapp.changeProjectApp</li> </ul>         |
| Chat                  | Chat with customers to resolve their service issues.                                               | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.chatService</li> <li>• <b>UI App Name:</b><br/>sap.crm.chatservice.uiapp.chatInteraction</li> </ul>                            |

| Business Service Name                | Description                                                                                                        | Business Service: Technical Details                                                                                                                                                                                               |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Code List Restrictions               | Restrict the availability of code values based on a combination of business entity, code field, and control field. | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.codeListRestrictionService</li> <li>• <b>UI App Name:</b><br/>sap.crm.codelistrestrictionsservice.uiapp.codeListRestrictionsUi</li> </ul>     |
| Competitor                           | Create new competitors and maintain competitor information.                                                        | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.md.service.competitorService</li> <li>• <b>UI App Name:</b><br/>sap.crm.md.competitorservice.uiapp.competitorApp</li> </ul>                           |
| Competitor Product                   | Create and maintain comprehensive records of competitor products.                                                  | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.md.service.competitorProductService</li> <li>• <b>UI App Name:</b><br/>sap.crm.md.competitorproductservice.uiapp.competitorProductAdminApp</li> </ul> |
| Computer Telephony Integration (CTI) | Speak with customers using partner-supplied computer telephony service.                                            | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.ctiService</li> <li>• <b>UI App Name:</b><br/>sap.crm.ctiservice.uiapp.cti</li> </ul>                                                         |
| Contacts                             | Maintain the details for contacts by creating a contact or using the relation between a contact and an account.    | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.md.service.contactPersonService</li> <li>• <b>UI App Name:</b><br/>sap.crm.md.contactpersonservice.uiapp.contactApp</li> </ul>                        |
| Contract Account                     | Manage contract account and their related plans.                                                                   | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.contractAccountService</li> <li>• <b>UI App Name:</b><br/>sap.crm.contractaccountservice.uiapp.contractaccout</li> </ul>                      |

| Business Service Name             | Description                                                                             | Business Service: Technical Details                                                                                                                                                                                                                                                                                                |
|-----------------------------------|-----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Customer Insights                 | Analyze customer data and feedback to generate insights.                                | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.insightsService</li> <li>• <b>UI App Name:</b><br/>sap.crm.insightsservice.uiapp.insightssap.crm.insightsservice.uiapp.insightsAdminApp</li> </ul>                                                                                             |
| Customer Insights Summary Service | Summarize customer data to enable sales teams to customize their approach               | Technical Name:<br>sap.crm.service.insightsSummaryService                                                                                                                                                                                                                                                                          |
| Customer Returns                  | Create, manage, and track return orders.                                                | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.returnOrderService</li> <li>• <b>UI App Name:</b><br/>sap.crm.returnorderservice.uiapp.returnOrderAdmin</li> </ul>                                                                                                                             |
| Data Connector Service            |                                                                                         | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.dataConnectorService</li> <li>• <b>UI App Name:</b><br/>sap.crm.dataconnector.service.uiapp.dataConnectorConfigAppsap.crm.dataconnector.service.uiapp.communicationSystemsApp<br/>sap.crm.dataconnector.service.uiapp.monitoringApp</li> </ul> |
| Data Import and Export            | Import and export data.                                                                 | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.dataImpexService</li> <li>• <b>UI App Name:</b><br/>sap.crm.dataimpexservice.uiapp.dataimpex</li> </ul>                                                                                                                                        |
| Data Depersonalization            | Remove personal data of employees, individual customers, and contacts on their request. | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.dppService</li> <li>• <b>UI App Name:</b><br/>sap.crm.dppservice.uiapp.depersonalize</li> </ul>                                                                                                                                                |
| Data Disclosure                   | Disclose personal data of employees, individual customers, and contacts.                | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.dppService</li> <li>• <b>UI App Name:</b><br/>sap.crm.dppservice.uiapp.disclosure</li> </ul>                                                                                                                                                   |

| Business Service Name                           | Description                                                                                                    | Business Service: Technical Details                                                                                                                                                                                                     |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data Workbench                                  | Import data from a third-party system or a csv file.                                                           | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.dataworkbenchService</li> <li>• <b>UI App Name:</b><br/>sap.crm.dataworkbenchservice.uiapp.dataworkbench</li> </ul>                                 |
| Digital Assistant - CX AI Toolkit Admin Service | Integrate the CX AI Toolkit from the settings page.                                                            | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.digitalAssistantCxAitkAdminService</li> <li>• <b>UI App Name:</b><br/>sap.crm.digitalassistantcxaitkadminservice.uiapp.cxaiToolKitAdmin"</li> </ul> |
| Digital Assistant - CX AI Toolkit Service       | Analyze enterprise data with trusted AI that helps you improve sales performance and enhance customer service. | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.digitalAssistantCxAitkService</li> <li>• <b>UI App Name:</b><br/>sap.crm.digitalassistantcxaitkservice.uiapp.cxaiToolKit</li> </ul>                 |
| Digital Selling Workspace                       | Gather information and insights required for productive customer conversations                                 | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.dswService</li> <li>• <b>UI App Name:</b><br/>sap.crm.dswservice.uiapp.dswApp</li> </ul>                                                            |
| Document Service                                | Configure document services.                                                                                   | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.documentService</li> <li>• <b>UI App Name:</b><br/>sap.crm.documentservice.uiapp.document</li> </ul>                                                |
| E-mail Action                                   | Add or remove e-mails                                                                                          | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.emailActionService</li> <li>• <b>UI App Name:</b><br/>sap.crm.emailactionservice.uiapp.emailAction</li> </ul>                                       |
| E-mail Channel                                  | Create and manage e-mail channels used for customer interactions and case routing.                             | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.emailChannelService</li> <li>• <b>UI App Name:</b><br/>sap.crm.emailchannelservice.uiapp.emailChannel</li> </ul>                                    |

| Business Service Name    | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Business Service: Technical Details                                                                                                                                                                        |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E-mail Monitor           | Monitor inbound and outbound e-mails.                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.emailMonitorService</li> <li>• <b>UI App Name:</b><br/>sap.crm.emailmonitorservice.uiapp.emailMonitor</li> </ul>       |
| E-mail MTA Configuration | Configure a Mail Transfer Agent (MTA) to send e-mails through your company's Simple Mail Transfer Protocol (SMTP) server or to use the default MTA.                                                                                                                                                                                                                                                                                                                                                       | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.emailMtaConfigService</li> <li>• <b>UI App Name:</b><br/>sap.crm.emailmtaconfigservice.uiapp.emailMtaConfig</li> </ul> |
| E-mail                   | Create and send e-mail messages to customers.                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.emailService</li> <li>• <b>UI App Name:</b><br/>sap.crm.emailservice.uiapp.email</li> </ul>                            |
| E-mail Interaction       | Track e-mail interactions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.emailService</li> <li>• <b>UI App Name:</b><br/>sap.crm.emailservice.uiapp.emailInteraction</li> </ul>                 |
| Employees                | Create employees in the system.                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.md.service.employeeService</li> <li>• <b>UI App Name:</b><br/>sap.crm.md.employeeservice.uiapp.employeeApp</li> </ul>          |
| Enrichment Service       | The enrichment service is used only by other services such as autoflow and e-mail templates to provide additional information about a related entity. For example: a case can be linked to partners, but the case only contains the partner ID. If more information about the partner is required, such as the partner name, a service like autoflow can call the enrichment service to retrieve the additional information. The enrichment service has no app, and isn't recommended for any user roles. | <b>Technical Name:</b><br>sap.crm.service.enrichmentService                                                                                                                                                |

| Business Service Name           | Description                                                                                                         | Business Service: Technical Details                                                                                                                                                                                     |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ESM Backend Integration Service | Enables an integration option with backend systems such as an S/4HANA system                                        | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.esmBackendIntegrationService</li> <li>• <b>UI App Name:</b><br/>sap.crm.esmbackendintegrationsevice.uiapp.integrationApp</li> </ul> |
| ESM Common Control Service      |                                                                                                                     | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.esmCommonControlService</li> <li>• <b>UI App Name:</b><br/>sap.crm.esmcommoncontrolservice.uiapp.commonControlApp</li> </ul>        |
| ESM Condition Service           |                                                                                                                     | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.esmConditionService</li> <li>• <b>UI App Name:</b><br/>sap.crm.esmconditionservice.uiapp.conditionApp</li> </ul>                    |
| Event Bridge                    | Enable event bridge service for your users.                                                                         | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.eventBridgeService</li> <li>• <b>UI App Name:</b><br/>sap.crm.eventbridgeservice.uiapp.eventBridgeAdmin</li> </ul>                  |
| Extensibility Admin             | Create extension fields and define the field attributes. You can also create custom validations and determinations. | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.extensibilityService</li> <li>• <b>UI App Name:</b><br/>sap.crm.extensibilityservice.uiapp.extensibilityAdmin</li> </ul>            |
| External ID Configuration       |                                                                                                                     | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.externalIdAdminService</li> <li>• <b>UI App Name:</b><br/>sap.crm.externalidadminservice.uiapp.externalIdAdminApp</li> </ul>        |

| Business Service Name | Description                                                                                                                                         | Business Service: Technical Details                                                                                                                                                                                                 |
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| External Link         |                                                                                                                                                     | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.externalNavigationService</li> <li>• <b>UI App Name:</b><br/>sap.crm.externalnavigation-service.uiapp.externalNavigationApp</li> </ul>          |
| Favorites             | Mark frequently used objects to enable quick access in the future.                                                                                  | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.favoriteService</li> <li>• <b>UI App Name:</b><br/>sap.crm.favoriteservice.uiapp.favorite</li> </ul>                                            |
| Favorites Workspace   | View all your favorites in one place.                                                                                                               | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.favoriteService</li> <li>• <b>UI App Name:</b><br/>sap.crm.favoriteservice.uiapp.favoriteWorkspace</li> </ul>                                   |
| Feedback              | Configure feedback service.                                                                                                                         | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.feedbackService</li> <li>• <b>UI App Name:</b><br/>sap.crm.feedbackservice.uiapp.feedbacksap.crm.feedbackservice.uiapp.feedbackAdmin</li> </ul> |
| Financials            | Create financial transactions and their related services for contract account.                                                                      | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.financialsService</li> <li>• <b>UI App Name:</b><br/>sap.crm.financialsservice.uiapp.financials</li> </ul>                                      |
| Flag                  |                                                                                                                                                     | <b>Technical Name:</b><br>sap.crm.service.flagService                                                                                                                                                                               |
| Flexibility           | Change the look and feel of the solution for all users by changing layout settings, adding mash-ups and fields, as well as adding extension fields. | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.extensibilityService</li> <li>• <b>UI App Name:</b><br/>sap.crm.extensibilityservice.uiapp.flex</li> </ul>                                      |
| Form                  | Configure the form service to work with forms.                                                                                                      | <b>Technical Name:</b><br>sap.crm.service.formService                                                                                                                                                                               |



| Business Service Name | Description                                                                                                                   | Business Service: Technical Details                                                                                                                                                                                                                                                                                  |
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| Functional Location   | Maintain physical location of the object that requires servicing or maintenance.                                              | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.functionalLocationService</li> <li>• <b>UI App Name:</b><br/>sap.crm.functionallocationservice.uiapp.functionalLocationsap.crm.functionallocationservice.uiapp.functionalLocationAdmin</li> </ul>                                |
| Global Calendar       | View all appointments or visits days of an organizational unit within a particular company or region.                         | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.calendarService</li> <li>• <b>UI App Name:</b><br/>sap.crm.calendarService.uiapp.globalCalendarApp</li> </ul>                                                                                                                    |
| Groupware Integration | Use SAP Sales Cloud Version 2 features from Microsoft Outlook or your Gmail inbox.                                            | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.groupwareSsiService</li> <li>• <b>UI App Name:</b> <ul style="list-style-type: none"> <li>• sap.crm.groupwaressiservice.uiapp.groupwareSsi</li> <li>• sap.crm.groupwaressiservice.uiapp.groupwareSsiAdmin</li> </ul> </li> </ul> |
| Guided Selling        | Work through the sales cycle across various phases by creating opportunities.                                                 | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.guidedSellingService</li> <li>• <b>UI App Name:</b><br/>sap.crm.guidedsellingservice.uiapp.guidedSelling</li> </ul>                                                                                                              |
| Holiday Calendar      | Add holidays of an organizational unit within a particular company or region.                                                 | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.holidayCalendarService</li> <li>• <b>UI App Name:</b><br/>sap.crm.holidaycalendarservice.uiapp.holidayCalendarApp</li> </ul>                                                                                                     |
| Home Page             | Explore the default start page that provides various card types and helps you with essential information and easy navigation. | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.homepageService</li> <li>• <b>UI App Name:</b><br/>sap.crm.homepageservice.uiapp.homepage</li> </ul>                                                                                                                             |

| Business Service Name                                | Description                                                                        | Business Service: Technical Details                                                                                                                                                                                              |
|------------------------------------------------------|------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Home Page Admin                                      | Arrange the start page with information and activities based on business roles.    | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.homepageService</li> <li>• <b>UI App Name:</b><br/>sap.crm.homepageservice.uiapp.homepageadmin</li> </ul>                                    |
| Identity and Access Management Configuration Service | Configure authentication and authorization mechanism for your users.               | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.iamConfigurationService</li> <li>• <b>UI App Name:</b><br/>sap.crm.iamconfiguration.service.uiapp.iamConfiguration</li> </ul>                |
| Inbound Data Connector                               |                                                                                    | <b>Technical Name:</b><br>sap.crm.service.inboundDataConnectorService                                                                                                                                                            |
| Individual Customers                                 | Manage, monitor, store, and track business information about individual customers. | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.md.service.individualCustomerService</li> <li>• <b>UI App Name:</b><br/>sap.crm.md.individualcustomer.service.uiapp.individualCustomerApp</li> </ul> |
| Installed Bases                                      | Arrange installed items hierarchically at your customer's location.                | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.installedBaseService</li> <li>• <b>UI App Name:</b><br/>sap.crm.installedbaseservice.uiapp.installedBase</li> </ul>                          |
| Knowledge Base                                       | Search for, review, and share Knowledge Base Articles.                             | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.knowledgeBaseService</li> <li>• <b>UI App Name:</b><br/>sap.crm.knowledgebaseservice.uiapp.knowledgebase</li> </ul>                          |
| Knowledge Base Settings                              | Configure knowledge base settings.                                                 | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.knowledgeBaseService</li> <li>• <b>UI App Name:</b><br/>sap.crm.knowledgebaseservice.uiapp.knowledgebaseAdmin</li> </ul>                     |

| Business Service Name                   | Description                                                                                                                 | Business Service: Technical Details                                                                                                                                                                                                                                                                                            |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Language Adaptation Service             | Override the standard SAP texts and customize field label terminology to fit your business needs.                           | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.languageAdaptationService</li> <li>• <b>UI App Name:</b><br/>sap.crm.languageadaptationService.uiapp.languageAdaptation</li> </ul>                                                                                                         |
| Language, Region, & Currency            | Choose language, region, and currency settings                                                                              | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.i18nService</li> <li>• <b>UI App Name:</b><br/>sap.crm.i18nService.uiapp.i18napp</li> </ul>                                                                                                                                                |
| Language, Region, & Currency Admin      | Configure language, region, and currency settings                                                                           | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.i18nService</li> <li>• <b>UI App Name:</b><br/>sap.crm.i18nService.uiapp.i18nAdmin</li> </ul>                                                                                                                                              |
| Lead                                    | Capture interest in products and services, and qualify and nurture the interest to convert them to opportunities for sales. | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.leadService</li> <li>• <b>UI App Name:</b> <ul style="list-style-type: none"> <li>• sap.crm.service.leadService (<a href="#">Lead</a>)</li> <li>• sap.crm.service.leadService (<a href="#">Lead Administration</a>)</li> </ul> </li> </ul> |
| Library                                 | Store, share files with the other users, and link files as attachments in sales and service entities.                       | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.libraryService</li> <li>• <b>UI App Name:</b> <ul style="list-style-type: none"> <li>• sap.crm.libraryservice.uiapp.libraryApp</li> <li>• sap.crm.libraryservice.uiapp.libraryAdminApp</li> </ul> </li> </ul>                              |
| Machine Learning Topic Analyzer Service | Use Generative AI to view trending topics.                                                                                  | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.mlNlpTopicAnalyzerService</li> <li>• <b>UI App Name:</b><br/>sap.crm.mlNlpTopicAnalyzerService.uiapp.topicAnalyzerResult</li> </ul>                                                                                                        |

| Business Service Name                             | Description                                                                                        | Business Service: Technical Details                                                                                                                                                                                                                                    |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Machine Learning Management Generative AI Service | Use Generative AI to generate account synopsis, case summary, and draft e-mail response for cases. | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/><code>sap.crm.service.mlScenarioManagementGenAIService</code></li> <li>• <b>UI App Name:</b><br/><code>sap.crm.mlscenariomanagementgenaiservice.uiapp.mlScenarioMngmntGenAIAdmin</code></li> </ul> |
| Machine Learning Management Service               | Use case text to identify sentiment, translate case text, and identify profane language.           | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/><code>sap.crm.service.mlScenarioManagementService</code></li> <li>• <b>UI App Name:</b><br/><code>sap.crm.mlscenariomanagementservice.uiapp.mlScenarioMngmntAdmin</code></li> </ul>                |
| Mashups                                           | Integrate SAP Cloud solution data with data from online Web services or applications.              | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/><code>sap.crm.service.mashupService</code></li> <li>• <b>UI App Name:</b><br/><code>sap.crm.mashupservice.uiapp.mashupApp</code></li> </ul>                                                        |
| Mashup Authoring                                  | Activate preconfigured mashups and create new mashups.                                             | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/><code>sap.crm.service.mashupService</code></li> <li>• <b>UI App Name:</b><br/><code>sap.crm.mashupservice.uiapp.mashupAuthoring</code></li> </ul>                                                  |
| Mashup Configuration                              | Configure, access, and use mashups.                                                                | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/><code>sap.crm.service.extensibilityService</code></li> <li>• <b>UI App Name:</b><br/><code>sap.crm.extensibilityservice.uiapp.flexMashupConfigAdmin</code></li> </ul>                              |
| Mashup Demo                                       | Display mashup service consumption.                                                                | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/><code>sap.crm.service.mashupService</code></li> <li>• <b>UI App Name:</b><br/><code>sap.crm.mashupservice.uiapp.mashupDemo</code></li> </ul>                                                       |
| Messaging                                         | Exchange text messages with customers to resolve their service issues.                             | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/><code>sap.crm.service.chatService</code></li> <li>• <b>UI App Name:</b><br/><code>sap.crm.chatservice.uiapp.messageInteraction</code></li> </ul>                                                   |

| Business Service Name                | Description                                                                                                    | Business Service: Technical Details                                                                                                                                                                                                                                                 |
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| Microsoft Teams Service              |                                                                                                                | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.msteamsService</li> <li>• <b>UI App Name:</b><br/>sap.crm.msteamsservice.uiapp.ms<br/>teamsadminAppsap.crm.msteamsser<br/>vice.uiapp.msteamssap.crm.mstea<br/>msservice.uiapp.mobile</li> </ul> |
| Mobile Admin Service                 | Configure mobile application for your users.                                                                   | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.mobileAdminServ<br/>ice</li> <li>• <b>UI App Name:</b><br/>sap.crm.mobileadminservice.uiap<br/>p.mobile</li> </ul>                                                                              |
| Move Process                         | Create move processes and their related serv-<br>ices.                                                         | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.moveIntegration<br/>Service</li> <li>• <b>UI App Name:</b><br/>sap.crm.moveintegrationservice.<br/>uiapp.moveApp</li> </ul>                                                                     |
| Note                                 | Store and maintain notes with rich text support<br>for host objects like case, account, visit, and so<br>on.   | <b>Technical Name:</b><br>sap.crm.service.noteService                                                                                                                                                                                                                               |
| Opportunity                          | Configure access restrictions for Guided Selling,<br>Pipeline Manager, Forecast Tracker, and Pipeline<br>Flow. | <b>Technical Name:</b><br>sap.crm.service.opportunityService                                                                                                                                                                                                                        |
| Open Resource Discov-<br>ery Service |                                                                                                                | <b>Technical Name:</b><br>sap.crm.service.openResourceDiscove<br>ryService                                                                                                                                                                                                          |
| Organizational Units                 | Create an element of an organizational model.                                                                  | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.md.service.organization<br/>alUnitService</li> <li>• <b>UI App Name:</b><br/>sap.crm.md.organizationalunitse<br/>rvice.uiapp.organizationalUnitA<br/>pp</li> </ul>                                      |

| Business Service Name              | Description                                           | Business Service: Technical Details                                                                                                                                                                                                                                |
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| Organizational Unit Admin          | Maintain organizational units.                        | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.md.service.organizationalUnitService</li> <li>• <b>UI App Name:</b><br/>sap.crm.md.organizationalunitse<br/>rvice.uiapp.organizationalUnitA<br/>dminApp</li> </ul>                     |
| Outbound Data Connector            |                                                       | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.outboundDataCon<br/>nectorService</li> <li>• <b>UI App Name:</b><br/>sap.crm.outbounddataconnectorse<br/>rvice.uiapp.eventMonitoringApp</li> </ul>                             |
| Output Management                  | Manage activities related to the output of documents. | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.outputManagemen<br/>tService</li> <li>• <b>UI App Name:</b><br/>sap.crm.outputmanagementservice<br/>.uiapp.outputManagementApp</li> </ul>                                      |
| Page Layouts and Assignments Rules | Create and assign rules to page layouts.              | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.pageLayoutServi<br/>ce</li> <li>• <b>UI App Name:</b><br/>sap.crm.pagelayoutservice.uiapp<br/>.assignmentRules<br/>sap.crm.pagelay<br/>outservice.uiapp.pageLayouts</li> </ul> |
| Partners                           |                                                       | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.md.service.partnerServi<br/>ce</li> <li>• <b>UI App Name:</b><br/>sap.crm.md.partnerservice.uiapp<br/>.partnerApp</li> </ul>                                                           |
| Partner Contact Service            |                                                       | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.md.service.partnerConta<br/>ctService</li> <li>• <b>UI App Name:</b><br/>sap.crm.md.partnercontactservic<br/>e.uiapp.partnerContactApp</li> </ul>                                      |

| Business Service Name | Description                                                                                                           | Business Service: Technical Details                                                                                                                                                                                                                                                                                                           |
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| Party Processing      | Configure and maintain involved parties in various business transactions.                                             | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.partyProcessingService</li> <li>• <b>UI App Name:</b><br/>sap.crm.partyprocessingservice.uiapp.partyprocessingservice.uiapp.partySchemeApp</li> </ul>                                                                                                     |
| Phone                 | Track and review phone call activities.                                                                               | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.phoneService</li> <li>• <b>UI App Name:</b><br/>sap.crm.phoneservice.uiapp.phone</li> </ul>                                                                                                                                                               |
| Playbook              | Build a sales framework to guide your sales representatives through the entire sales cycle and help accelerate deals. | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.salesPlaybookService</li> <li>• <b>UI App Name:</b><br/>sap.crm.salesplaybookservice.uiapp.salesPlaybook</li> </ul>                                                                                                                                       |
| Premise               | Manage premise and their related services.                                                                            | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.premiseService</li> <li>• <b>UI App Name:</b><br/>sap.crm.premiseservice.uiapp.premise</li> </ul>                                                                                                                                                         |
| Pricing               | Configure pricing components and master data to enable native price calculation.                                      | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.pricingMasterDataService</li> <li>• <b>UI App Name:</b> <ul style="list-style-type: none"> <li>• sap.crm.pricingmasterdataservice.uiapp.pricingMasterData</li> <li>• sap.crm.pricingmasterdataservice.uiapp.pricingMasterDataAdmin</li> </ul> </li> </ul> |
| Products              | Create, view, and update products in an organizational master data.                                                   | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.md.service.productService</li> <li>• <b>UI App Name:</b><br/>sap.crm.md.productservice.uiapp.productApp</li> </ul>                                                                                                                                                |

| Business Service Name         | Description                                                                      | Business Service: Technical Details                                                                                                                                                                                                                                                                                                     |
|-------------------------------|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Product Groups                | Create, view, and update product groups in an organizational master data.        | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.md.service.productGroupService</li> <li>• <b>UI App Name:</b><br/>sap.crm.md.productgroupservice.uiapp.productGroupApp</li> </ul>                                                                                                                           |
| Product List                  | Configure targeted product proposals, exclusions, listings.                      | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b> business services<br/>sap.crm.service.productListService</li> <li>• <b>UI App Name:</b> <ul style="list-style-type: none"> <li>• sap.crm.productlistservice.uiapp.productListApp</li> <li>• sap.crm.productlistservice.uiapp.productListAdminApp</li> </ul> </li> </ul> |
| Product Installation Point    | Maintain physical location of the object that requires servicing or maintenance. | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.productInstallPointService</li> <li>• <b>UI App Name:</b><br/>sap.crm.productinstallpointservice.uiapp.productInstallPointsap.crm.productinstallpointservice.uiapp.productInstallPointAdmin</li> </ul>                                              |
| Qualtrics XM Discover Service | Create and configure feedback flow.                                              | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.qualtricsXmDiscoverService</li> <li>• <b>UI App Name:</b><br/>sap.crm.qualtricsxmdiscoverservice.uiapp.xmDiscoverAppLauncher</li> </ul>                                                                                                             |
| Question Bank                 | Store survey questions in the question bank for easy reuse in multiple surveys.  | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.questionBankService</li> <li>• <b>UI App Name:</b><br/>sap.crm.questionbankservice.uiapp.questionBankApp</li> </ul>                                                                                                                                 |
| Read Access Logging Service   | Monitor and record access to sensitive personal data.                            | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.md.service.raAppService</li> <li>• <b>UI App Name:</b><br/>sap.crm.md.raappservice.uiapp.raApp</li> </ul>                                                                                                                                                   |



| Business Service Name | Description                                                                                  | Business Service: Technical Details                                                                                                                                                                                                                                                                           |
|-----------------------|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Registered Products   | Use products that are associated with a specific customer and generally has a serial number. | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.registeredProductService</li> <li>• <b>UI App Name:</b><br/>sap.crm.registeredproductservice.uiapp.registeredProduct</li> </ul>                                                                                           |
| Roles                 | Configure different user roles in the system to access specific services.                    | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.roleService</li> <li>• <b>UI App Name:</b><br/>sap.crm.roleservice.uiapp.roleAdmin</li> </ul>                                                                                                                             |
| Campaign              | Engage with customers and prospects to promote products and services.                        | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.campaignService</li> <li>• <b>UI App Name:</b><br/>sap.crm.service.campaignService</li> </ul>                                                                                                                             |
| Sales Order           | Record and view sales interactions related to sales orders.                                  | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.salesOrderService</li> <li>• <b>UI App Name:</b> <ul style="list-style-type: none"> <li>• sap.crm.salesorderservice.uiapp.salesOrderAdmin</li> <li>• sap.crm.salesorderservice.uiapp.salesOrderApp</li> </ul> </li> </ul> |
| Sales Quote           | Create and manage sales quotations for business partners.                                    | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.salesQuoteService</li> <li>• <b>UI App Name:</b> <ul style="list-style-type: none"> <li>• sap.crm.salesquoteservice.uiapp.salesQuoteApp</li> <li>• salesquoteservice.uiapp.salesQuoteAdminApp</li> </ul> </li> </ul>      |
| SAML IdP Service      | Configure SAML IdP services for your users.                                                  | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.saml2idpService</li> <li>• <b>UI App Name:</b><br/>sap.crm.saml2idpservice.uiapp.idpConfigAdmin</li> </ul>                                                                                                                |

| Business Service Name             | Description                                              | Business Service: Technical Details                                                                                                                                                                                        |
|-----------------------------------|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Search                            | Search for entity details across the solution.           | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.searchService</li> <li>• <b>UI App Name:</b><br/>sap.crm.searchservice.uiapp.searchApp</li> </ul>                                      |
| Search Admin                      | Manage global search settings.                           | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.searchService</li> <li>• <b>UI App Name:</b><br/>sap.crm.searchservice.uiapp.searchAdmin</li> </ul>                                    |
| Search Referring Service          | Maintain the attributes of referring entities up to date | <b>Technical Name:</b><br>sap.crm.service.searchReferringService                                                                                                                                                           |
| Security Policies                 | Create security policies for your organization           | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.securityPolicyService</li> <li>• <b>UI App Name:</b><br/>sap.crm.securitypolicy.service.uiapp.securityPolicyAdmin</li> </ul>           |
| Service Category Catalog          | Configure service catalog classifications.               | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.serviceCategoryService</li> <li>• <b>UI App Name:</b><br/>sap.crm.servicecategoryservice.uiapp.serviceCatalogClassification</li> </ul> |
| Service Category Catalog Settings | Create and manage service category catalogs.             | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.serviceCategoryService</li> <li>• <b>UI App Name:</b><br/>sap.crm.servicecategoryservice.uiapp.serviceCatalogAdmin</li> </ul>          |
| Service Category Path             |                                                          | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.serviceCategoryService</li> <li>• <b>UI App Name:</b><br/>sap.crm.servicecategoryservice.uiapp.serviceCategoryPath</li> </ul>          |

| Business Service Name              | Description                                                 | Business Service: Technical Details                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------------|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Service Category Catalog Selection |                                                             | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.serviceCategoryService</li> <li>• <b>UI App Name:</b><br/>sap.crm.servicecategoryservice.uiapp.serviceCatalogSelection</li> </ul>                                                                                                                                                                   |
| Service Level                      | Create, view, and maintain service level agreements.        | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.serviceLevelService</li> <li>• <b>UI App Name:</b><br/>sap.crm.servicelevelservice.uiapp.serviceLevel</li> </ul>                                                                                                                                                                                    |
| Service Orders                     | Replicate service orders to external systems.               | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.s4hcServiceOrderService (cloud)</li> <li>• sap.crm.service.s4hopServiceOrderService(On-prem mashup)</li> <li>• <b>UI App Name:</b><br/>sap.crm.s4hcserviceorderservice.uiapp.externalReplica (cloud)</li> <li>• sap.crm.s4hopserviceorderservice.uiapp.s4hopServiceOrder(On-prem mashup)</li> </ul> |
| Service Quotation                  | Create and manage service quotations for business partners. | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.s4hcServiceQuotationService</li> <li>• <b>UI App Name:</b><br/>sap.crm.s4hcservicequotationservice.uiapp.s4hcServiceQuotation</li> <li>• sap.crm.servicequotationservice.uiapp.serviceQuotationAdmin</li> </ul>                                                                                     |
| Settings                           | Configure dashboard settings for your system.               | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.adminService</li> <li>• <b>UI App Name:</b><br/>sap.crm.adminservice.uiapp.admin</li> </ul>                                                                                                                                                                                                         |

| Business Service Name  | Description                                                                            | Business Service: Technical Details                                                                                                                                                                                                                                                          |
|------------------------|----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Settings               | Configure administration settings for your system.                                     | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.adminService</li> <li>• <b>UI App Name:</b><br/>sap.crm.adminservice.uiapp.adminconsole</li> </ul>                                                                                                       |
| Shell Service          | Adjust the interface to optimize it for interaction.                                   | <b>Technical Name:</b><br>sap.crm.service.shellService                                                                                                                                                                                                                                       |
| Surveys - Design Time  | Create surveys that sales representatives can use to collect business-related data.    | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.surveyDesignService</li> <li>• <b>UI App Name:</b><br/>sap.crm.surveydesignservice.uiapp.survey</li> </ul>                                                                                               |
| Surveys - runtime      | Answer Surveys and help collect important information for deriving insights.           | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.surveyService</li> <li>• <b>UI App Name:</b><br/>sap.crm.surveydesignservice.uiapp.survey</li> </ul>                                                                                                     |
| Surveys Administration | Configure your settings for surveys.                                                   | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/>sap.crm.service.surveyDesignService</li> <li>• <b>UI App Name:</b><br/>sap.crm.surveydesignservice.uiapp.surveyAdminApp</li> </ul>                                                                                       |
| Target Group           | Segment customers and prospects to form the right target audience for sales campaigns. | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b> <ul style="list-style-type: none"> <li>• sap.crm.service.targetGroupService</li> <li>• sap.crm.service.targetGroupMemberService</li> </ul> </li> <li>• <b>UI App Name:</b><br/>sap.crm.service.targetGroupService</li> </ul> |

| Business Service Name | Description                                                                                                | Business Service: Technical Details                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------|------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tasks                 | Create and manage your tasks associated with entities such as leads, opportunities, and quotes, and so on. | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b> <code>sap.crm.service.taskService</code></li> <li>• <b>UI App Name:</b> <ul style="list-style-type: none"> <li>• <code>sap.crm.taskservice.uiapp.taskApp</code></li> <li>• <code>sap.crm.taskservice.uiapp.taskAdmin</code></li> <li>• <code>sap.crm.taskservice.uiapp.taskTile</code></li> <li>• <code>sap.crm.taskservice.uiapp.taskManagerApp</code></li> </ul> </li> </ul> |
| Templates             | Create and edit templates.                                                                                 | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b> <code>sap.crm.service.templateService</code></li> <li>• <b>UI App Name:</b> <code>sap.crm.templateservice.uiapp.template</code></li> </ul>                                                                                                                                                                                                                                     |
| Template Settings     | Configure template settings.                                                                               | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b> <code>sap.crm.service.templateService</code></li> <li>• <b>UI App Name:</b> <code>sap.crm.templateservice.uiapp.templateAdmin</code></li> </ul>                                                                                                                                                                                                                                |
| Timeline              | View timeline events for your apps.                                                                        | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b> <code>sap.crm.service.timelineService</code></li> <li>• <b>UI App Name:</b> <code>sap.crm.timelineservice.uiapp.timeline</code></li> </ul>                                                                                                                                                                                                                                     |
| Timeline Settings     | Configure timeline for your apps.                                                                          | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b> <code>sap.crm.service.timelineService</code></li> <li>• <b>UI App Name:</b> <code>sap.crm.timelineservice.uiapp.timelineAdmin</code></li> </ul>                                                                                                                                                                                                                                |
| User Settings         | Configure user settings.                                                                                   | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b> <code>sap.crm.service.userService</code></li> <li>• <b>UI App Name:</b> <code>sap.crm.userservice.uiapp.userAdmin</code></li> </ul>                                                                                                                                                                                                                                            |

| Business Service Name    | Description                                                                                                                   | Business Service: Technical Details                                                                                                                                                                                                                |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Utilities Admin Settings | Configure utilities admin settings.                                                                                           | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/><code>sap.crm.service.commonIntegrationService</code></li> <li>• <b>UI App Name:</b><br/><code>sap.crm.commonintegrationsevice.uiapp.commonadminui</code></li> </ul>           |
| Value Mapping            |                                                                                                                               | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/><code>sap.crm.service.valueMappingService</code></li> <li>• <b>UI App Name:</b><br/><code>sap.crm.valuemappingservice.uiapp.valuemappingApp</code></li> </ul>                  |
| Warranty                 | Assign a warranty to a registered product or installation point, and determine its coverage in a ticket.                      | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/><code>sap.crm.service.warrantyService</code></li> <li>• <b>UI App Name:</b><br/><code>sap.crm.warrantysevice.uiapp.warranty</code></li> </ul>                                  |
| Where-Used Service       | Get a list of content, within a system, where objects such as extensions fields, code lists, and so on is used or referenced. | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/><code>sap.crm.service.whereUsedService</code></li> <li>• <b>UI App Name:</b><br/><code>sap.crm.whereusedsevice.uiapp.whereUsed</code></li> </ul>                               |
| Working Day Calendar     | Add working days of an organizational unit within a particular company or region.                                             | <ul style="list-style-type: none"> <li>• <b>Technical Name:</b><br/><code>sap.crm.service.workingDayCalendarService</code></li> <li>• <b>UI App Name:</b><br/><code>sap.crm.workingdaycalendarsevice.uiapp.workingDayCalendarApp</code></li> </ul> |

## 3.3 Security Policies

A security policy is a set of rules that defines password complexity, such as including numerical digits and password validity, like requiring a password change after a certain period of time.

You can define multiple security policies because work areas or departments of a company may have different password security requirements.

As an administrator, you can manage the security level, if desired, by editing and enhancing the security policy, for example, by changing the complexity and validity for all passwords, in accordance with your company's security requirements.

## 3.3.1 Security Policy Features

Lists and describes the features of a security policy.

The security policy features are classified as:

- System password-related features
  - Enable Password Login - if enabled, allows users to log in to the system with a password.
  - Set Default - if enabled is set as the default security policy for all the users.
- System password complexity-related features
  - Minimum Number of Characters - defines the minimum number of characters in a valid password.
  - Minimum Number of Lowercase Alphabets - defines the minimum number of lowercase alphabets in a valid password.
  - Minimum Number of Uppercase Alphabets - defines the minimum number of uppercase alphabets in a valid password.
  - Minimum Number of Digits - defines the number of numerals in a password.
  - Minimum Number of Special Characters - defines the minimum number of special characters in a valid password.
- System password validity-related features
  - Allowed Number of Incorrect Password Attempts - defines the number of successive login failures before a password is locked. A message is displayed on the login screen once the password is locked.
  - Password History - defines the number of passwords to store in the memory to compare with the new productive password set by the user.
  - Maximum Password Validity (in Days) - defines the maximum number of days a password is valid for.
  - Minimum Waiting Time Before Changing Password (in Days) - defines the minimum number of days a user must wait before changing the existing password to a new password.
  - Unused Productive Password Validity (in Days) - defines the validity of the productive password after changing from the initial password.

## 3.3.2 Create Security Policies

You can define multiple security policies as different work areas or departments in a company can have different password security requirements.

### Context

As an administrator, you can increase the security level by editing and enhancing a security policy. For example, by changing the complexity and validity for all passwords, in accordance with your company's security requirements.

## Procedure

1. Log in to the system as an administrator.
2. Navigate to [User Menu](#) on the top-right corner, and access the [System Settings](#) page.
3. Go to [All Settings](#) > [Users and Control](#) > [Security Policies](#).
4. Click the add icon (+) on the [Security Policies](#) page.
5. Enter a [Policy ID](#), [Policy Name](#), and [Description](#) for the security policy. You can further choose to set this policy as the default using the [Set Default](#) switch.
6. Enhance security complexity for your policy by entering inputs for: [Minimum Number of Characters](#), [Minimum Number of Lowercase Alphabets](#), [Minimum Number of Uppercase Alphabets](#), [Minimum Number of Digits](#), [Minimum Number of Special Characters](#), [Allowed Number of Incorrect Password Attempts](#), [Password History](#), [Maximum Password Validity \(in Days\)](#), [Minimum Password Change Waiting Time \(in Days\)](#), and then choose [Save](#).
7. Refresh the [Security Policies](#) page to view your newly created security policy. All the security policies that you've created appear in the list format on the [All Policies](#) page.

You've now created a security policy.

### 3.3.3 Edit Security Policy

Administrators can edit security policies.

## Procedure

1. Log in to the system as an administrator.
2. Navigate to [User Menu](#) on the top-right corner, and access the [System Settings](#) page.
3. Go to [All Settings](#) > [Users and Control](#) > [Security Policies](#).
4. Click an existing policy ID to view the details. The [Security Policy](#) quick view opens.
5. Hover the mouse over the fields that you want to edit. Click on pencil icon next to the field to edit the field.

## Results

Once edited, the changes are saved automatically.



## 3.4 Identity Provider Configuration

Identity provider refers to an entity that manages identity information and provides authentication services to trusted service providers.

Every user type must authenticate to the system for regular browser-based front-end access, as well as for electronic data exchange, such as business-to-business communication. The system doesn't support anonymous access. We recommend using Single Sign-On (SSO) for basic security. To protect accounts further, configure the identity provider (IdP) of the SSO solution to provide enhanced security.

### 3.4.1 Configure Identity Provider

Administrators can configure identity provider to enable Single-Sign-On (SSO) for the system.

#### Enable Single Sign-On for the System

To enable SSO for a tenant, you've to connect the identity provider to the tenant. To do so, download the metadata of your system, upload the tenant metadata XML file to your identity provider (IdP), and then download the configured metadata from your identity provider. Afterwards on your system, choose the identity provider, and create the identity provider to start using the system.

##### Download the System Metadata

Download the tenant metadata XML and further upload it to IdP to configure it.

1. Log in to the system as an administrator.
2. Navigate to the [User Menu](#) on the top-right corner, and access the [System Settings](#) page.
3. Go to ► [All Settings](#) ► [Users and Control](#) ► [IdP Configuration](#) ► page.
4. Choose [Download Metadata](#). The tenant metadata XML file is downloaded to your local drive.

##### Note

If your IdP certificate is due for renewal in the next 20 days, the message "Click the [Renew and Download Metadata](#) button before the certificate validate expiration date to avoid disruptions to your SSO users" appears. In addition to the message, the certificate validity period is displayed as well.

If you intend to renew and reconfigure your identity provider, click the [Renew and Download Metadata](#) option to download and further reconfigure the identity provider. The [Renew and Download Metadata](#) option extends the validity of your certificate and stops your SSO users from logging into the system until you've reconfigured your IdP with the renewed metadata file. Use the downloaded service-provider metadata file to configure your IdP.

#### Configure System Metadata in Your Identity Provider (IdP)

Upload the system metadata XML file to IdP, configure it, and then download the configured IdP metadata file.

### Note

Use this procedure to configure authentication services provided by SAP Cloud Identity Services. You can follow similar steps to configure other IdPs as well.

1. Log on to the [SAP Cloud Identity Services](#) as an administrator. You can also use an external identity provider.
2. On the SAP Cloud Identity Services screen, navigate to [Applications & Resources > Applications](#), and choose [Add](#) to create a new application for your system.
3. In the [Add Application](#) pop-up window, enter a name, and choose [Save](#). A new application page opens.
4. Under [Trust > Single Sign-On](#), configure the following settings:
  - Choose [Type](#), and select SAML 2.0
  - Choose [SAML 2.0 Configuration](#), and upload the downloaded tenant metadata XML from your local drive.
  - Choose [Subject Name Identifier](#), and configure the attribute, which the application uses to identify the users.
5. Navigate to [Tenant Settings](#), and choose [SAML 2.0 Configuration](#) to open a new screen.
6. Choose [Download Metadata File](#) to download the IdP metadata file.

### Create a Trusted Identity Provider and Upload IdP Metadata to Your System

Navigate to your tenant to create a trusted IdP and choose a default IdP.

1. Log in to your system as an administrator.
2. Navigate to the [User Menu](#) on the top-right corner, and access the [System Settings](#) page.
3. Go to [All Settings > Users and Control > Configure IdP](#).
4. Choose the add (+) icon under the [Trusted Identity Provider](#) section.
5. Under the [Upload IdP Metadata](#) quick view, enter an [Alias](#), and upload the configured metadata XML file (downloaded from your IdP) by either dropping or browsing to upload from the system.
6. Choose [Save](#) to save your configuration.
7. Activate your configuration by enabling the [Set Active](#) switch. To make it a default option, turn on the [Set Default](#) switch.

You've now configured Single Sign-On for your system.

## Related Information

[Access Admin Console](#)

## 3.5 System for Cross-Domain Identity Management API

System for Cross-domain Identity Management (SCIM) specification is designed to efficiently manage user identities in cloud-based applications and services.

The specification suite builds upon the existing schema and deployments, placing specific emphasis on simplicity of development and integration, while applying existing authentication, authorization, and privacy

models. It intends to reduce the cost and complexity of user management operations by providing a common user schema and extension model, as well as binding documents to provide patterns for exchanging this schema using standard protocols.

## Getting Started with the SAP Sales and Service Cloud V2 SCIM API

The SAP Sales and Service Cloud V2 SCIM API is an API based on the SCIM (System for Cross-domain Identity Management) specification. It's used as an interface between SAP Cloud Identity Services (IAS) and SAP Sales and Service Cloud V2 solutions to transfer user information as part of user management. In SAP Sales and Service Cloud V2, the SCIM API is used to provision Users and User Groups.

- User Service: [https://api.sap.com/api/SalesSvcCloudV2\\_scimUserService/overview](https://api.sap.com/api/SalesSvcCloudV2_scimUserService/overview)
- User Group Service: [https://api.sap.com/api/SalesSvcCloudV2\\_scimGroupService/overview](https://api.sap.com/api/SalesSvcCloudV2_scimGroupService/overview)

SAP Sales and Service Cloud V2 have a standalone user and roles management. To efficiently manage the users and user groups in V2 in an integrated scenario, SCIM API is implemented.

### → Remember

- The incoming data from the Source (SCIM API) is considered as master data and overwrites any User attributes already maintained in the SAP Sales and Service Cloud Version 2 system.
- Each group has 1:1 mapping with the business role in SAP Sales and Service Cloud Version 2.
- SCIM API is applicable for management of business users but not to integration users (the users that are created using communication systems).
- To integrate and consume the SCIM API, create a technical API user.
- Username is the unique attribute and is used to resolve any conflicts.
- SAP Sales and Service Cloud Version 2 don't support hard-deletion or soft-deletion of a user. The user status is set to Obsolete when a Delete request is received.
- When a create user request is received with an existing Username, a 409 – CONFLICT response is raised, and the user can be updated with a PUT request.
- When a Create request is received with an existing deleted user (Status: Obsolete), the same user is reactivated with the same ID.
- When a create Group request is received with an existing Display Name (Role Name), the request isn't rejected but is linked to the existing role. If the role isn't existing, a new Role is created with Default Services assigned.
- When you use a PUT request to update the Members list, the system removes the role from any existing users in SAP Sales and Service Cloud Version 2 who aren't included in the updated list.
- If it's required to add or remove roles for specific users, PATCH operation on the Group Resource can be used to modify the members.
- Whenever a role is created in SAP Sales and Service Cloud Version 2, it's automatically available as a User Group when the groups' endpoint is accessed.
- The default security policy assigned: **S\_BUSINESS\_USER\_WITHOUT\_PASSWORD**.

## Create a Technical API User

You can consume the SCIM APIs using the Technical User (API User). To create a technical user:

1. Navigate to the [User Menu](#) on the top-right corner, and access the [System Settings](#) page.
2. Go to ► [All Settings](#) ► [Users and Control](#) ► [Roles](#) ►.
3. Select [Technical Roles](#) from the dropdown, and create a new technical role as an admin role. Assign the [User](#), [Role](#), [Employee](#), and [Admin Service](#) services.
4. Save and activate the role.
5. Navigate to ► [All Settings](#) ► [Users and Control](#) ► [Users](#) ►.
6. Select [Technical Users](#) from the dropdown, create a new technical user, and assign the previously created technical role to this user.

## Authentication Mechanism

SCIM API supports the following authentication mechanisms:

- Basic Authentication
- Certificate Authentication
- Bearer Token Authentication

## Monitoring Tools

SAP Sales and Service Cloud Version 2 doesn't provide any monitoring tools for the direct usage of SCIM APIs. As a user, you must monitor the replications using SCIM APIs.

## 3.6 Settings

This page lists all the identity and access management settings.

### Cross-Site Request Forgery Settings

Cross-Site Request Forgery (CSRF) is a web-security vulnerability that allows an attacker to induce users to perform actions that they don't intend to perform. CSRF attacks exploit the trust a Web application has in an authenticated user. To prevent such attacks, CSRF settings are provided.

Configure Cross-Site Request Forgery Settings

1. Log in to the system as an administrator.

2. Navigate to the [User Menu](#) on the right-top corner, and access the [System Settings](#) page.
3. Go to ► [All Settings](#) ► [Users and Control](#) ► [Settings](#) ► [Cross-Site Request Forgery Settings](#) ►.
4. Select one of the modes: [Strict](#), [None](#), or [Lax](#) from the dropdown for third-party integration.

#### 📘 Note

- [Lax](#): This mode allows cookies to be sent for top-level navigation only.
- [None](#): This mode disables SameSite-based protection. The website can use its own CSRF protection mechanisms.
- [Strict](#): This mode disables cookies being sent to all third-party websites. Cookies are sent only if the domain is the same as the path for which the cookie has been set.

#### 📘 Note

By default, the [Strict](#) mode is enabled for third-party integrations. However, for existing customers, this change doesn't affect the existing integrations in the current release. In a future release, the existing integrations can be affected.

## 4 Security Settings

As an administrator, you can define security settings that are applicable for all users.

### 4.1 File and Attachment Processing

Define the allowed file types for attachments and discover how to handle temporary files.

#### 4.1.1 Configure MIME Types

As an administrator, you can define the file types that can be uploaded to your solution.

The Multipurpose Internet Mail Extensions (MIME) type configuration controls the files you can add to the SAP Service Cloud Version 2 system. These file types include attachment uploads as well as files sent via e-mail attachments.

You can upload attachment files to your solution in several application scenarios, for example, in billing, in data migration, or image files of your travel expense receipts. Regularly updated anti-virus software checks the uploaded files for viruses and other types of malicious software. Note that file-name extensions can be changed to disguise the actual file format of the file.

We recommend that you start with a minimal MIME list, as you've the option of adding more later. Choose from the list of allowed MIME types for uploading documents that are specific for your project.

Follow these steps to select MIME types from the provided list:

1. Log on as an administrator with access rights to the MIME Type settings
2. From your user profile, navigate to ► [Settings](#) ► [All Settings](#) ► [Data Administration](#) ► [Allowed MIME Types for Attachment Upload](#) ►
3. In the new screen, add and allow MIME types relevant to your project.

#### ⚠ Caution

If you enable additional MIME types, exercise caution, as different file types can increase the security risks. For example, although it's possible to allow EXE files, we recommend that you refrain from enabling them.

#### 4.1.2 Temporary Files

Your browser saves temporary files as you work. Use your browser tools to delete cached information.

The stored data is currently limited to static files such as, meta data, UI layout, java script files and files.

### → Recommendation

The solution doesn't delete these types of temporary entries. To remove cached data, we recommend using the appropriate features of your browser.

## 4.2 Set Auto Sign Out

As an administrator, you can set the sign out time for all users or for a selected business role.

### Context

For security reasons, users are automatically logged off from the system if they've been inactive in the system for a certain period. If you leave this option empty, inactive users will be logged off from the system after 1 hour.

### Procedure

1. Navigate to ► [User Menu](#) ► [System Settings](#) ► [All Settings](#) ►.
2. Under [System Preferences](#), select [Appearance and Behavior](#).
3. Go to the [Behavior](#) tab.
4. Under the [Auto Sign Out](#) section, set the sign out time for the following:
  - Company
  - Business Users
5. To set the sign out time for all the inactive users in the company, select the dropdown and choose any time period.
6. To set the sign out time for a specific inactive business users, click **+** available under [Business Role](#) to create a new row.
7. Search and select a business role, choose a time, and click **✓** to create.  
Similarly, you can add many business roles as required.
8. To delete any of the rows, click **✕**.
9. Save your changes.

You can lose your unsaved changes in the following scenarios:

- **Browser Inactivity**  
If your SAP Sales and Service Cloud V2 browser tab becomes inactive (for example, if you switch to another tab and stop interacting with the application), your session may time out. One minute before auto sign out, you'll see a prompt asking if you want to stay signed in. If you don't respond, you will be automatically logged out.

#### Note

If you have multiple tabs of the application open, inactive tabs may be logged out even if you are active in another tab.

- **Logged in on Multiple Devices**  
If you are signed in to the same account on more than one device (such as a laptop and a tablet), inactivity is tracked separately on each device. Being active on one device does not prevent a session from timing out on another.
- **Network Disconnection**  
If your network connection is lost, for example, if you lock your system or close your laptop, you will be logged out when the connection is restored. You must log in again.
- **System Shutdown**  
If your device shuts down, any unsaved changes will be lost.

To use the SSO option when you sign in, use the [Sign in using SSO](#) option. If you have a direct integration with the SAP Sales and Service Cloud V2 site, append the URL with `auth=sso`. For example, `https://<tenant-url>?auth=sso`. This parameter skips the log in page.

## 4.3 Content Security Policy

A Content Security Policy (CSP) is a security standard that provides an additional layer of protection from cross-site scripting (XSS), clickjacking, and other code injection attacks.

The solution ships with default settings to provide additional security protections. For enhancements and integrations, you can modify the settings or add additional settings if needed. You must, however, be careful if you want to change the [Unsafe Eval](#) or [Unsafe Inline](#) settings since these can disable or severely weaken the protections.



# 5 Generative AI

When using Generative AI to generate content, such as account summaries, case summaries, or draft email responses, ensure that you follow certain security measures.

Some of the measures are as follows:

- **GenAI Output Verification:** You must verify the results generated by GenAI (such as case summaries), as these outputs may require manual validation.
- **Personal Data Handling:** There is a possibility that personal data (PII) could be included in GenAI processes. To address this, you must use the PII removal features when necessary.

## Related Information

[Generative AI](#)

# 6 Logs and Changes

A record of events that occur within the solution.

Logs and changes are essentially text-based files or streams that capture important information about the software's execution, including errors, warnings, informational messages, user interactions, system status updates, and performance metrics.

The benefits of logs and changes in your solution include the following:

- Efficient trouble shooting and issue resolution
- Performance optimization
- Real-time monitoring to avoid security threads
- Use of compliance and auditing purposes
- Effective security and historical analysis

[Audit Logs \[page 58\]](#)

Review security-relevant chronological records of user actions.

[Changes \[page 60\]](#)

Review detailed configuration changes with effective filtering capability.

[Deletions \[page 62\]](#)

Review the details of deleted entities.

## Related Information

[Depersonalization Logs \[page 70\]](#)

View the summary of all removal runs in the [Depersonalization Logs](#) worklist.

## 6.1 Audit Logs

Review security-relevant chronological records of user actions.

### Context

Audit logs help monitor important changes in the SAP Sales and Service Cloud V2 system. This includes changes to configuration settings, modifications to critical data, and unexpected behaviors.

## Procedure

1. Navigate to [User Menu](#) > [System Settings](#) > [All Settings](#).
2. Under [Logs and Changes](#), select [Audit Logs](#).

A detailed list of audit logs opens. There are two tabs to help you view the audit logs based on the time.

- **Recent:** Logs up to seven days are displayed in this tab.
- **Archived:** Logs older than 7 days are displayed here.

You can define the retention period of the logs from 7 to 14 months. By default, it is considered as 7 months.

Download both recent and archived logs using [Export](#).

You can download archived data for up to 30 days at a time. Downloading of files is allowed only once in 24 hours.

Messages and data in the downloaded audit logs are Base64 encoded. Each record in the log has an associated event type denoted by a number based on the following table:

Event Type Code

| Code | Event                  |
|------|------------------------|
| 0    | Login                  |
| 1    | Logout                 |
| 2    | Login failed           |
| 3    | User created           |
| 4    | User updated           |
| 5    | Password updated       |
| 6    | Configuration changed  |
| 7    | Personal data accessed |
| 8    | Personal data modified |
| 10   | Authorization denied   |

To verify the integrity of the downloaded file, there is an additional column [Checksum](#) which is calculated using SHA-256. The result is a hexadecimal string of ([id](#) + [timestamp](#) + [eventType](#) + [sourceIp](#) + [message](#) + [service](#) + [tenantId](#) + [userId](#) + [data](#)).

### Note

The following conditions should be considered for calculating the checksum.

- [eventType](#) is integer.
- For archived logs [Message](#) and [Data](#) are first decoded from Base64.
- Only the first 50 characters of [Message](#) are considered.

The event type [Authorization Granted](#) appears only under [Recent](#), and will not be archived. For future use, you can download the logs from [Recent](#).

3. In the [Recent](#) tab, use the filters based on time, event type, and service. In the [Archived](#) tab, use the date range filter for accurate data.

4. To get more information about a specific log entry, click  (Show Data) available in the [Data](#) column.

## 6.2 Changes

Review detailed configuration changes with effective filtering capability.

### Context

Change history refers to a record of alterations made to business entities over time. It captures details such as when changes were made, who made them, and what specific modifications occurred. This history provides a chronological account of revisions, ensuring transparency, accountability, and the ability to track the evolution of data or content. Unlike Audit Logs, this feature provides more information regarding the configuration changes.

From the list view of [Changes](#), you can select IDs and users' names, where navigation is available, to open them in quick view. Select



(Open in Detail View) to open them in a new tab with more details.

This feature displays records of changes for the following entities:

- Accounts
- Account Hierarchy
- Activity Assignment Rules
- Activity Plans
- Appointments
- Call Lists
- Campaign
- Cases
- Change Projects
- Chats
- Competitors
- Competitor Products
- Contract Accounts
- Delivery Points
- Emails
- Functional Locations
- In-Person Interactions
- Installed Base
- Installation Points
- Key Account Plans

- Leads
- Letter Interactions
- Partners
- Partner Contacts
- Phone Calls
- Plants
- Premise
- Product Installation Points
- Contact Person
- Employee
- Individual Customers
- Organizational Unit
- Products
- Product Groups
- Product Lists
- Opportunities
- Registered Products
- Return Orders
- Roles
- Route Groups
- Sales Orders
- Sales Playbook
- Sales Quotes
- Sales Territories
- Service Catalogs
- Service Categories
- Service Orders
- Service Quotations
- Target Groups
- Tasks
- Users
- Visits

#### → Remember

The list of change history includes data based on your access restrictions to these entities.

## Procedure

1. Navigate to ► [User Menu](#) ► [System Settings](#) ► [All Settings](#) ►.
2. Under [Logs and Changes](#), select [Changes](#).

3. Change the type of the data storage based on the following:

- [Recent Data](#): To view data up to six months ago
- [Older Data](#): To view data older than six months. This data will be deleted after three years.

#### ⚠ Restriction

The attributes filter is only available for recent data. Older data is stored differently and the attributes information isn't available for filtering.

4. Select a business entity from the drop-down menu.

5. Use the other filter options such as [Changed By](#), [Changed On](#), and so on, to get the accurate data in the list.

6. Optional: Use the following options based on your requirements:

- To export the list of changes, select  (Export). Choose the file type and then select [Export](#). The system displays the progress in the [Task Progress](#) section. Once the status changes to [Completed](#), you can select the task to download it.
- To search for changes, use  (Search).
- To refresh the list, select  (Refresh).

#### 📘 Note

- View the same information in all the supported business entities by navigating to the [Administrative Data](#) tab.
- When an entity is deleted, all change history entries for this entity are also deleted.
- When a data subject gets depersonalized, their corresponding change history records are also deleted.

## Related Information

[Data Protection and Privacy \[page 66\]](#)

Use Data Protection and Privacy to manage personal and sensitive personal data of employees, individual customers, and contacts. As an employee responsible for data protection and privacy regulation compliance in an organization, you can disclose as well as remove data on request.

## 6.3 Deletions

Review the details of deleted entities.

### Context

It captures details such as when the entities were deleted, who deleted them, and so on.

You can select users' names, where navigation is available, to open them in quick view. Select



(Open in Detail View) to open them in a new tab with more details.

This feature displays records of deletions for the following entities:

- Account Hierarchy
- Activity Assignment Rules
- Activity Plans
- Appointments
- Call Lists
- Campaigns
- Change Projects
- Chats
- Emails
- Functional Locations
- Installed Base
- Leads
- Letter Interactions
- Opportunities
- Person Interactions
- Phone Calls
- Plants
- Premises
- Product Groups
- Product Installation Points
- Products
- Registered Products
- Roles
- Route Groups
- Sales Orders
- Sales Playbooks
- Sales Quotes
- Sales Territories
- Service Orders
- Target Groups
- Tasks
- Users
- Visits

## Procedure

1. Navigate to ► [User Menu](#) ► [System Settings](#) ► [All Settings](#) ►.

2. Under *Logs and Changes*, select *Deletions*.
3. Select a business entity from the drop-down menu.
4. Use the other filter options such as *Deleted By* and *Deleted On* to get the accurate data in the list.



# 7 Front-End Security

The frontend is the main gate to your web application and is open to your users and clients. Therefore, it's vital that you have security measures in place to protect your users.

The SAP Cloud Native Stack front end consists of Web application user interfaces that support the following features:

- X-Frame-options response header to avoid clickjacking attacks
- Cross-site request forgery (CSRF) protection
- Secure channels (HTTPS/TLS1.2)
- Cookie-based authentication using a double-submit cookie
- Angular framework protections, UI and domain protections for URL mashups.
- Access to business data only after authentication and with sufficient authorizations using identity management and Role-Based Access Management (RBAM).
- Cross-site-scripting counter measures such as Content Security Policy (CSP)
- Strict Transport Security (STS) header

## 8 Data Protection and Privacy

Use Data Protection and Privacy to manage personal and sensitive personal data of employees, individual customers, and contacts. As an employee responsible for data protection and privacy regulation compliance in an organization, you can disclose as well as remove data on request.

Data processing systems store master data or transactional data used to perform business processes and to document them. In many cases, it involves the personal data of employees, individual customers, and contacts. In many countries/regions, the storage, disclosure, and removal of such personal data from data storage systems must be in accordance with statutory data protection laws. One requirement in many countries/regions is that the personal data can only be stored if a clear business reason for this data retention exists. Most data protection legislation proscribes fixed retention periods, defining how long data can be stored in data systems, after which it must be deleted. In addition, legislation in many countries/regions stipulates that the data protection officer must disclose the personal data of individuals, when they expressly request it.

### Note

In this document, employees, individual customers, and contacts are collectively referred to as data subjects.

## Features

There are a number of key features of Data Protection and Privacy. These are outlined as follows:

### **Data Disclosure — Obligation to Disclose**

A key principle in data protection and privacy is the Obligation to Disclose. This is an obligation set in legislation in many countries/regions where data protection regulation has been adopted. As an administrator responsible for data protection regulation compliance, you can disclose personal data of employees, individual customers, and contacts. You can display a summary of all data associated with these business partners stored in the system. You can also access the detailed records.

### **Data Removal — Deletion on Request**

This second data protection and privacy principle refers to the requirement of organizations to delete personal data held on its business partners that is kept in an identifiable form, and retain this data for no longer than necessary. Where specified, organizations must delete all such personal data after the relevant data retention period has elapsed.

### **Data Protection Roles**

In large organizations, employees with the designated role (Data Protection Officer, for example) are responsible for ensuring that data protection and privacy principles are followed, and that the organization complies with all data protection and privacy legislation in force within the country/region (or countries/regions) it operates. However, these tasks can be delegated to other authorized employees, for example, designated Human Resources administrators.

### **Authorization**

The [Data Disclosure](#) screen is only available to authorized employees or Data Privacy officers in your organization. It's therefore recommended this service assignment is only given to those employees directly responsible for data protection and privacy regulation compliance in your organization.

#### Note

Employees, such as Data Protection officers with responsibility for data protection have full access rights for the [Data Disclosure](#) screens. Because of the ability for an individual user to access large volumes of personal employee data across many business entities, the access log is provided to allow transparency and traceability of user access to personal data. The log doesn't contain detailed personal data, but rather a summary of the types of data accessed, when, and by whom it was accessed.

## 8.1 Data Disclosure

Disclose personal data of employees, individual customers, contacts, and supplier contacts in the [Data Disclosure](#) worklist.

As an administrator responsible for data protection regulation compliance, you can disclose personal data of employees, individual customers, and contact. You can display a summary of all data associated with these data subjects stored in the system. You can also access the detailed records.

#### Note

In this document, employees, individual customers, contacts, and supplier contacts are collectively referred to as data subjects.

### Procedure

1. Navigate to your user profile on the top-right corner of your screen, and from the dropdown list select  [Settings](#)  [All Settings](#)  [Data Protection and Privacy](#)  [Data Disclosure](#) . The system opens a new tab.
2. To display the disclosure data for employees, individual customers, contacts, and supplier contacts, select the relevant option from the dropdown. For example, if you want to disclose an employee's data, select [All Employees](#)
3. Click  ([Access Data](#)) next to the relevant data subject, in the [Actions](#) column.

#### Note

A dialog box appears informing you that your access to this screen is logged. Confirm this message to proceed.

The system runs a background job to retrieve the data, and generates a **Run ID**.

4. To view all the details of the accessed data, click [Disclosure Runs](#) and then click the [Run ID](#) of the data subject whose data you want to access.
5. In the new window, under the [Disclosable Data](#) header you can view all individual records that can be disclosed. Click  ([Expand](#)) to expand and view individual data record summaries for the selected entity.

6. Click the links for the individual records, for example, [General Data](#) or transactional data, such as [Cases](#) to navigate to the actual data record held in the system.

#### Note

The figure shown in the [No. of Records](#) column represents the number of discreet data records (for example, Cases) of the selected type assigned to the employee in the SAP Service Cloud Version 2 system. A zero indicates that no records of this type exist for the selected employee.

You can also view all the disclosable data records together by clicking  ([Preview](#)). To download disclosable data for the given data subject click  ([Download](#)) .

#### Note

Currently only .json file type is supported for downloading data.

7. Close the overview tab to return to the [Data Disclosure](#) worklist.

You've successfully viewed a summary of all personal data that can be disclosed to an individual who requests it.

#### Note

To view the summary of all data subjects accessed, click  (View Logs) under [Actions](#). The system maintains the logs for a period of 201 days, after which they're automatically deleted from the database. If you want to define a different retention period for the disclosure logs, navigate to the [Retention Rule for Logs](#) screen, and under the [Retention Period](#) column hover your mouse to view the  ([Edit](#)) icon. Click the icon and enter the desired retention period.

## 8.2 Data Depersonalization

Remove personal data of employees, individual customers, contacts, and supplier contacts on their request in the [Data Depersonalization](#) worklist.

You must remove personal data (e.g. business partners, transactions) once they reach their end of purpose. SAP Service Cloud Version 2 offers business partner driven removal that deletes the person and all the related data/transactions.

It is now possible for you, as the person responsible for data protection regulation compliance, to delete personal data of employees, individual customers, and contacts on their request.

#### Note

In this document, employees, individual customers, contacts, and supplier contacts are collectively referred to as data subjects.

## Prerequisite

You've defined the retention periods relevant for your country/region in the [Data Retention Rule for Employees](#) worklist. If you configure this rule, the system runs a query for the employee validity dates against the retention period. An employee can be removed only if they've passed the retention period.

## Procedure

1. Navigate to your user profile on the top-right corner of your screen, and from the dropdown list select [Settings](#) > [All Settings](#) > [Data Protection and Privacy](#) > [Data Depersonalization](#). The system opens a new tab.
2. To display data for removal of employees, individual customers, contacts, and supplier contacts, select the relevant option from the drop-down. For example, if you want to remove an employee's data, select [All Employees](#).  
You can also view the list of all employees that have passed the retention period by selecting [Employees Past Retention Period](#) from the dropdown.
3. Select the **Remove** icon (🗑️) in the [Action](#) column next to the relevant data subject. The system runs a check to ensure that there are no retention blocks or vetoes raised by any other entities. If there are no blocks or vetoes, the system runs a background job to delete the data. You can view the [Run ID](#) in the [Depersonalization Logs](#) worklist.
4. Optional: You also have an option to block the removal for a data subject. This may be required for business or legal reasons. As long as the data subject is blocked, it cannot be removed. To do this, click the **Block Depersonalization** icon (🔒) against the data subject. The system prompts you to enter the reason and description for blocking. Clicking [OK](#) blocks the removal of this data subject. To unblock a data subject, click the **Unblock Depersonalization** icon (🔓) next to the data subject.
5. Close the overview tab to return to the [Data Depersonalization](#) worklist.

## Related Information

[Data Retention \[page 71\]](#)

If the purpose for which the data you acquired is not valid anymore, but you must retain it for audit purposes, you can export the data before deleting it from the system.

### 8.2.1 Depersonalization Behavior

A summary of the depersonalization behavior of all data subject for the different transaction types.

Depersonalization can happen in the following ways:

- **Manual Trigger:**  
When an individual customer is depersonalized, it signals the end of the business with that particular customer. In this context, there's no longer a purpose to retain any related information. The deletion

or depersonalization process proceeds unless there are vetoes from ongoing transactions (for example, a current business activity or an ongoing quote) or retention period blockers. In the absence of such constraints, all relevant data is slated for deletion or depersonalization.

- **Automated trigger:**  
Background jobs systematically scan the system for transactions that have become outdated and delete them based on predefined configurations. Over time, customers without active business connections cease to be linked to any transactions. A secondary background job identifies customers without transactions and no recent changes, applying deletion or depersonalization procedures based on configuration settings. This automated process ensures the removal of stale customer data from the system.

#### Depersonalization Behavior

| Transaction Type | Behavior                                                                                                                                                                                                                                                           |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phone call       | If there's no veto, the transactions owned by the depersonalized business partner (employee, contact, or individual customer) are deleted.                                                                                                                         |
| Chat             | If there's no veto, the transactions owned by the depersonalized business partner (employee, contact, or individual customer) are deleted.                                                                                                                         |
| Letter           | If there's no veto, the transactions owned by the depersonalized business partner (employee, contact, or individual customer) are deleted.                                                                                                                         |
| In-Person        | If there's no veto, the transactions owned by the depersonalized business partner (employee, contact, or individual customer) are deleted.                                                                                                                         |
| Change history   | Records that belong to the depersonalized entity are deleted.                                                                                                                                                                                                      |
| Timeline         | Records that belong to the person being depersonalized are also deleted.                                                                                                                                                                                           |
| Case             | A case can raise a veto until the case is complete. When the employee assigned to the case is depersonalized, the associated notes and attachments are deleted. No other fields are anonymized on the case as employee data is retrieved from the employee itself. |

## 8.3 Depersonalization Logs

View the summary of all removal runs in the [Depersonalization Logs](#) worklist.

The following information is available in the logs:

- **Run ID**  
Click the [Run ID](#) to view the details of the entities from which the data subject was removed and the status along with timestamp.
- **Removal date**
- **User who initiated the removal**

- Data subject type
- Status

#### 📘 Note

These logs are maintained for a period of 201 days, after which they're automatically deleted from the database. If you want to define a different retention period for the depersonalization logs, navigate to the [Retention Rule for Logs](#) screen, and under the [Retention Period](#) column hover your mouse to view the [✎ \(Edit\)](#) icon. Click the icon and enter the desired retention period.

## 8.4 Read Access Logging

Use read access logging to monitor read-access to sensitive personal data. You can track who has accessed critical information and when.

SAP identifies and delivers the following attributes as sensitive personal:

- individualCustomerIdentification [identificationId]
- emailInteraction [htmlContent]
- emailInteraction [plainContent]

These attributes are enabled by default. Whenever a user views them, the system creates a Read Access Log (RAL) entry.

The following information is available in the logs:

- Channel
- Business Entity
- Business Entity ID
- Timestamp
- User who accessed

#### 📘 Note

The logs are maintained for a period of 14 days, after which they're automatically deleted from the database. Please raise a support ticket to request an export log of the data.

## 8.5 Data Retention

If the purpose for which the data you acquired is not valid anymore, but you must retain it for audit purposes, you can export the data before deleting it from the system.

You can request the export of the logs using a support ticket.

## Data Retention Rule for Employees

The retention period determines how much time must pass before the employee can be depersonalized. It is added to the employee validity end date and checked against the current date. Only after the retention period is over, it is possible to depersonalize the employee.

To define the employee retention periods, follow these steps:

1. Navigate to your user profile on the top-right corner of your screen, and from the dropdown list select [Settings](#) > [All Settings](#) > [Data Protection and Privacy](#) > [Retention Rule for Employees](#). The system opens a new tab.
2. In the [Minimum Retention Period](#) column, hover your mouse to view the **Edit** icon (✎).
3. Click ✎ and define the time you want the data to be retained for each country/region.
4. Click [Save](#).

## 8.6 Automate Removal of End of Purpose Transactions

As an administrator responsible for data protection regulation compliance, you can schedule automated deletion of transactions that have reached their end of purpose.

In the [Automated Transaction Removal](#) worklist, you can create a batch job to schedule deletion runs. You can schedule the runs immediately, or set a recurrence to continuously purge obsolete transactions from the system. The system selects all the transactions that have been set as obsolete before a certain cut-off date. Once selected, the system creates one data removal run per service.

### Procedure

1. Navigate to the user menu on the top-right corner of your screen, and from the dropdown list select [Settings](#) > [All Settings](#) > [Data Protection and Privacy](#) and Choose [Automated Transaction Removal](#). The system opens a new tab.

The [Service Name](#) column shows the business services for which the transactions can be removed automatically. Choose the service name link to open a quick view showing the [Retention Configuration](#) specific to that service.

Each business service can establish its own retention criteria, based on which the respective transactions are qualified for removal. The applications determine the required number of parameters in the retention criteria and specifies which parameters should be included. You can use these parameters to create your retention criteria.

For Example, Opportunity service allows combining duration (transaction age) with statuses. As a result, opportunities older than six months and marked as **Completed** or **Canceled** are eligible for removal.

Based on the retention criteria for each service, a background job runs daily at 12:00 AM (12-hour clock) or 00:00 (24-hour clock) UTC, identifying all qualifying transactions and marking them as End of Purpose (EOP).

The following table provides the retention configuration and parameters for each business service. It also gives an overview of the services that can either be deleted or depersonalized.



## Retention Configuration Parameters

| Business Service          | Parameters       | Delete | Depersonalize |
|---------------------------|------------------|--------|---------------|
| Appointments              | Duration, Status | Yes    |               |
| Call Lists                | Duration, Status | Yes    |               |
| Campaigns                 | Duration, Status | Yes    |               |
| Case                      | Duration         |        | Yes           |
| Chat Interaction          | Duration         | Yes    |               |
| Email Interaction         | Duration         | Yes    |               |
| Forecast                  | Duration         | Yes    |               |
| Leads                     | Duration, Status | Yes    |               |
| Opportunity               | Duration, Status | Yes    |               |
| Phone Interaction         | Duration         | Yes    |               |
| Promotion                 | Duration, Status |        |               |
| Relationship Intelligence | Duration         |        |               |
| Sales Orders              | Duration         |        | Yes           |
| Sales Quotes              | Duration         |        | Yes           |
| Signal                    | Duration         |        |               |
| Target Group              | Duration, Status | Yes    |               |
| Tasks                     | Duration, Status | Yes    |               |

- Choose  in the **Job Schedule** column for the service whose schedule you want to view. If there are no existing jobs, the system opens the **Create Schedule** quick view.
- Enter a description.
- Choose one of the following options:
  - Immediate:** Sets a one time job that runs at the given start date/time.
  - Recurring:** Sets a recurring job that runs at regular frequency. This results in multiple deletion runs for a single job schedule.
- Enter the **Date Offset** period. This acts as a buffer while removing all transactions that are marked EOP. For example, if the date offset is set as 10 days, then the transactions that are marked EOP today, will not be deleted in the next 10 days. They will be picked up for deletion by the job running after that.  
The data offset values in deletion job schedules should adhere to the following:
  - Minimum Value: 0
  - Maximum Value: 999 for days/months/years
  - Default Value: 99
- Choose **OK**.

To view the logs of the End of Purpose calculation runs, choose **End of Purpose Runs** and then choose  in the **Logs** column. The system opens a new tab with the log details.

To view the logs of the deletion runs, choose  in the **Deletion Runs** column. The new tab provides information such as the Run ID and status of runs. Choose  under the **Logs** column. The system opens a new tab with the log details of the deletion runs for the selected service. The new tab provides the detailed log for each run.

There may be multiple Run IDs for each job schedule depending on whether it is a recurring job or not.

### Note

Summary of status of the job runs:

- **READY:** Indicates that a run is scheduled but not yet started
- **TRIGGERED:** Indicates that the run has started
- **CANCELED:** Indicates that the run is canceled due to a technical error

## 8.7 Automate Removal of Business Partners

As an administrator responsible for data protection regulation compliance, you can schedule automated deletion of obsolete business partners, such as, employees, individual customers, contacts, and supplier contacts.

### Prerequisites

The business partners are already marked as obsolete.

### Context

In the **Automated Business Partner Removal** view, you can create a batch job to schedule deletion runs. You can schedule the runs immediately, or set a recurrence to continuously purge obsolete business partners from the system. The system selects all the business partners that have been set as obsolete before a certain cut-off date.

### Procedure

1. Navigate to the user menu on the top-right corner of your screen, and from the dropdown list select ► [Settings](#) ► [All Settings](#) ► [Data Protection and Privacy](#) ► and click **Automated Business Partner Removal** . The system opens a new tab.
2. Click  in the [Job Schedule](#) column next to the business partner for whom you want to create a run. The system opens the [Create Schedule](#) dialog.
3. Enter a description.
4. Choose if you want to schedule the run immediately or set a recurring run.
  - If you want to run it immediately, choose the date and time
  - If you want to set a recurring run, set the frequency of the run
5. Select the [Date Offset](#) period. This means that the business partners are removed from the system after the offset time is over, for example, 30 days after they are set to obsolete.

The data offset values in deletion job schedules should adhere to the following:

- Minimum Value: 0
- Maximum Value: 999 for days/months/years
- Default Value: 99

6. Optional:

To include business partners that are mapped to other systems, turn on the *Include Business Partners with ID Mapping* switch.

If you do not turn the switch on, the system excludes the business partners that are replicated in other external systems such as SAP S/4HANA, and ERP, and only triggers removal for business partners available locally in the system.

If you want the system to consider only those business partners that are marked as End of Purpose in the master data, select the *Only Business Partners marked as End of Purpose* switch. This ensures that the system does not pick up business partners that are not flagged.

7. Click *OK*

To view the logs of the deletion runs, click  in the *Deletion Runs* column. In the new tab, click  under the *Logs* column. The system opens a new tab that displays the status of all the individual removal runs for each business partner, and the corresponding Run ID, if already scheduled.

## 8.8 Automate End of Purpose for Business Partners

As a data protection officer, you can schedule end of purpose for business partners, such as contact persons, employees, and individual customers.

### Context

You can schedule a run to find business partners that are no longer used by application services. When the run is over, the system automatically updates the status of business partners to *Obsolete* and flags them to have reached their *End of Purpose*.

The business partners that have reached end of purpose can no longer be involved in new transactional documents.

#### Caution

This action is irreversible, and the business partner can no longer be viewed.

In the *Automated Business Partner End of Purpose Runs* view, you can create a batch job to schedule end of purpose runs. You can schedule the runs immediately or set a recurrence to continuously set the status of the business partners to *End of Purpose*. The system selects all the business partners that have not been used before a certain cut-off date. Once the selection is done and the status is set, the system removes the business partner data during the *Automated Business Partner Removal* run.

## Note

During the run, only local business partners are considered.

## Procedure

1. Navigate to the user menu on the top-right corner of your screen, and from the dropdown list select  [Settings](#)  [All Settings](#)  and click [Business Partner End of Purpose Runs](#).

The system opens a new tab displaying the different types of business partners such as Employee, Contact, and Individual Customer. You can schedule jobs or view results of scheduled jobs.

2. Choose a [Business Partner Type](#).
3. Click [Job Schedule](#) icon to open the [Create Deletion Job](#) popup.
4. Enter a [Run Description](#).
5. In the [Run Details](#) section, select one of the two: Immediate or Recurring.
6. If you select [Immediate](#), select a start date and time for the job.
7. If you select [Recurring](#), select a frequency for the job.
8. Enter the [Date Offset](#) period.

In the case of contact persons and individual customers, if you choose the offset date to [1 Month\(s\)](#), it means the business partner must have been changed over one month ago. Here, the system checks against the last changed date.

In the case of employees, when it is set to [1 Month\(s\)](#), it means the employee must have been terminated over one month ago. Here, the system checks against the termination date.

## Caution

Ensure you don't create recurrent schedules with high frequency.

9. Execute a simulation run with the [Simulation Mode](#) switch turned on. Turn off the [Simulation Mode](#) switch, if you want to schedule a productive run. It is advised to execute a simulation run before a real-time end of purpose run.

By using the simulation mode, you can verify the number of business partners for which the status is going to be changed, prevent the status of the business partners from changing, and see any errors that can occur during the run.

10. Select [OK](#).
11. To view the results of the job runs, go to [Business Partner End of Purpose Runs](#) screen, select [Job Runs](#) for a business partner type.

The screen displays details such as run ID, creation time, status and logs.

12. Select [Logs](#) to open the screen with the details of your run.

View details such as timestamps, process names, service names, created by, and created on. In the [Description](#) tab, the system displays the number of business partners for which the status has been changed, and the corresponding Run ID, if already scheduled.

### Note

#### Summarized Messages

**Message 1:** Simulation run scheduled for 10235 Business Partners of type Contact Person.

**Meaning 1:** This indicates the number of business partners picked up for the run.

**Message 2:** Number of Business Partners that are still referenced: 6441

**Meaning 2:** The number of business partners that are active and referenced in transactions.

**Message 3:** Number of Business Partners where end of purpose is set: 3794

**Meaning 3:** The number of business partners that could be updated.

## 8.9 Glossary

The following terms are general to SAP products. Not all terms may be relevant for this SAP product.

| Term                   | Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Blocking</b>        | A method of restricting access to data for which the primary business purpose has ended.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Consent</b>         | The action of the data subject confirming that the usage of his or her personal data shall be allowed for a given purpose. A consent functionality allows the storage of a consent record in relation to a specific purpose and shows if a data subject has granted, withdrawn, or denied consent. The legal, contractual, or in other form justified reason for the processing of personal data to complete an end-to-end business process. The personal data used to complete the process is predefined in a purpose, which is defined by the data controller. The process must be defined before the personal data required to fulfill the purpose can be determined. |
| <b>Data subject</b>    | An identified or identifiable natural person, defined in relation to applicable data protection legislation, for example, the EU GDPR.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Deletion</b>        | Deletion of <b>personal data</b> The legal, contractual, or in other form justified reason for the so that the data is no longer available.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>End of business</b> | Defines the end of active business and the start of residence time and retention period.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

| Term                       | Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| End of purpose (EoP)       | End of purpose and start of blocking period. The point in time when the primary processing purpose ends, for example, a contract is fulfilled.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| End of purpose (EoP) check | A method of identifying the point in time for a data set when the processing of <b>personal data</b> is no longer required for the primary <b>business purpose</b> . After the <b>EoP</b> has been reached, the data is <b>blocked</b> and can only be accessed by users with special authorization, for example, tax auditors.                                                                                                                                                                                                                                                                                                |
| Personal data              | Any information relating to an identified or identifiable natural person ("data subject"), defined in relation to applicable data protection legislation, for example, the EU GDPR.                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Purpose                    | The information that specifies the reason and the goal for the processing of a specific set of personal data. As a rule, the purpose references the relevant legal basis for the processing of personal data.                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Residence period           | The period of time between the end of business and the end of purpose (EoP) for a data set during which the data remains in the database and can be used in case of subsequent processes related to the original purpose. At the end of the longest configured residence period, the data is blocked or deleted. The residence period is part of the overall retention period.                                                                                                                                                                                                                                                 |
| Retention period           | The period of time between the end of the last business activity involving a specific object (for example, a business partner) and the deletion of the corresponding data, subject to applicable laws. The retention period is a combination of the residence period and the blocking period.                                                                                                                                                                                                                                                                                                                                  |
| Sensitive personal data    | <p>A category of personal data that usually includes the following type of information:</p> <ul style="list-style-type: none"> <li>• Special categories of personal data, such as data revealing racial or ethnic origin, political opinions, religious or philosophical beliefs, trade union membership, genetic data, biometric data, data concerning health or sex life or sexual orientation.</li> <li>• Personal data subject to professional secrecy</li> <li>• Personal data relating to criminal or administrative offenses</li> <li>• Personal data concerning insurances and bank or credit card accounts</li> </ul> |

| Term                                        | Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Technical and organizational measures (TOM) | <p>Some basic requirements that support data protection and privacy are often referred to as technical and organizational measures (TOM). The following topics are related to data protection and privacy and require appropriate TOMs, for example:</p> <ul style="list-style-type: none"> <li>• <b>Access control:</b> Authentication features</li> <li>• <b>Authorizations:</b> Authorization concept</li> <li>• <b>Read access logging</b></li> <li>• <b>Transmission control / Communication security</b></li> <li>• <b>Input control / Change logging</b></li> <li>• <b>Availability control</b></li> <li>• <b>Separation by purpose:</b> Is subject to the organizational model implemented and must be applied as part of the authorization concept.</li> </ul> |
| Where-used check (WUC)                      | <p>A process designed to ensure data integrity in the case of potential blocking of business partner data. An application's where-used check (WUC) determines if there is any dependent data for a certain business partner in the database. If dependent data exists, this means the data is still required for business activities. Therefore, the blocking of business partners referenced in the data is prevented.</p>                                                                                                                                                                                                                                                                                                                                             |

## 9 Security-Relevant Logging, Monitoring, and Alerting

Find information about security-relevant logging, monitoring, and alerting.

### 9.1 Infrastructure Audit Logs

All Amazon Web Services (AWS) accounts created for Cloud Native Scenarios are prepared by a central SAP team. AWS CloudTrail and connections to SAP's central monitoring center are set up by default.

The SAP Cyber Fusion Center monitors the collected logs and triggers alerts in case suspicious events are detected.

### 9.2 Application Security Audit Logs

- Security logs capture events, such as, failed authentication.
- The system maintains the logs for a period of 201 days, after which they're automatically deleted from the database.
- You can request the export of the logs using a support ticket.



# 10 Application Security

The solution provides features to prevent unauthorized access.

## 10.1 Cookies

In this section, you can find a list of cookies and their functions.

The application uses the following cookies to exchange information between the client and server. This information may include session IDs, load-balancing information, or performance indicators, for example.

To protect the information contained in the cookies, the system requires secure communication channels (HTTPS) and sets the `Secure` and `HttpOnly` flags for all cookies.

| Cookie                                     | Purpose       | When Set                       |
|--------------------------------------------|---------------|--------------------------------|
| <code>__Host-x-sap-crm-cookie</code>       | Session ID    | Created after successful logon |
| <code>__Secure-x-sap-refresh-cookie</code> | Refresh token | Created after successful logon |

# 11 Mashup Security Guidelines

This document outlines key security considerations, best practices, and known limitations when implementing mashups and web components in SAP Sales Cloud Version 2 and SAP Service Cloud Version 2. It is intended for developers, administrators, and customers integrating external content via mashups.

## General Mashup Security Guidelines

To ensure safe use of mashups in SAP Sales Cloud Version 2 and SAP Service Cloud Version 2:

- **Enforce HTTPS Protocol:** Always use HTTPS URLs in mashups to prevent browser warnings and ensure secure content loading.
- **Implement whitelisting for the Same-Origin Policy:** Browsers enforce the same-origin policy to restrict cross-domain access. Whitelisting third-party content is essential to maintain security boundaries in mashups.
- **Secure URL Mashups:** When creating mashups, ensure that the embedded content does not introduce vulnerabilities and complies with your organization's security policies.
- **Implement Authentication Mechanisms:** Ensure that data transmitted between CNS and external services is encrypted and that authentication mechanisms are in place to prevent unauthorized access.
- **Review HTML/Web component Mashup Content:** Before implementing HTML/Web component mashups, confirm that the embedded content complies with your organization's security and data privacy standards.
- **Configure Single Sign-On (SSO):** Implement SSO in mashups to enhance user experience and security, reducing the need for users to manage multiple credentials.
- **Monitor Third-Party Service Usage:** Be aware that third-party services integrated via mashups may monitor API usage; review their licensing and usage policies accordingly.
- **Review Mashup Configurations Periodically:** Regularly assess mashup configurations to ensure ongoing compliance with security standards and organizational policies.

## Input Validation for Mashup Parameters

### Current Validation:

- Name of parameters must be a **string**.
- Maximum input length is **80 characters**.

### Note

Only trusted users should have access to mashup parameter authoring. Customers are responsible for managing access controls appropriately.

## Loading External JavaScript Files via Web Component Mashup

Web components are reusable, encapsulated custom HTML elements that promote modularity and reusability in web development. You can integrate your personalized web components into applications using mashup authoring.

To implement a web component mashup, you must specify the following:

- The **resource URL** where the web component is hosted (a JavaScript file),
- The **custom element name** (for example, `<elix-calendar-month>`)
- The required **parameters/attributes** for configuring the web component within the mashup.

The component is embedded inside an iframe, isolating it from the core Sales and Service application and reducing the risk of conflicts or unintended behavior.

### Security Enforcement

- Mashups allow external web components only from trusted platforms by enforcing a Content Security Policy (CSP).
- The CSP ensures that external resources such as JavaScript files are loaded only from trusted domains. If a resource originates from an untrusted domain, the CSP blocks it, preventing the external file from loading.
- This security mechanism helps protect the application from malicious or unauthorized content by strictly controlling which external sources are permitted.

## CORS Configuration Risks and Mitigation

### Security Risks of Incorrect CORS:

- **Unauthorized Data Access:** Improperly configured CORS can allow malicious external domains to access sensitive data within SAP C4C, leading to potential data breaches.
- **Cross-Site Scripting (XSS) Attacks:** Embedding untrusted content without proper validation can expose the system to XSS attacks, compromising user sessions and data integrity.
- **Bypassing Same-Origin Policy:** Incorrect CORS settings might inadvertently disable browser-enforced same-origin policies, allowing unauthorized cross-domain requests.
- **Exposure of Sensitive APIs:** Misconfigured CORS can unintentionally expose internal APIs to external domains, increasing the attack surface.

### Recommended Best Practices:

- **Restrict Allowed Origins:** Specify only trusted domains in the CORS configuration to prevent unauthorized access.
- **Use HTTPS Protocol:** Ensure all mashup URLs use HTTPS to maintain data encryption and integrity during transmission.
- **Validate External Content:** Implement strict validation for any external content or scripts included in mashups to prevent injection attacks.
- **Regularly Review CORS Settings:** Periodically audit and update CORS configurations to adapt to changing security requirements and prevent misconfigurations.
- **Use CSP Appropriately:** Use content security policy to appropriately allow mashup content to be embedded in CNS screens.

## Single Sign-On (SSO) Configuration for Mashups

### Setup Guidelines:

- **Configure SSO** : Set up SSO for for SAP Sales Cloud V2 and SAP Service cloud V2 first. Refer to the **Related Information** section for detailed instructions. The same IAS must also be configured for third-party mashup content.
- **Review IAS restrictions**: Some IAS may have restrictions for embedding inside mashups which internally used iframe element. These restrictions are configured on the IAS side rather than Mashup.
- **Validate External Content**: Implement strict validation for any external content or scripts included in mashups to prevent injection attacks.
- **Use Mashup with SSO**: Embedding mashups with SSO follows the same steps as those without authentication. Ensure that the third-party web application in the mashup is configured with SSO using the same IDP as the SAP Sales and Service V2 tenant.

Mashups provide powerful integration capabilities but must be handled with strict security controls. Follow the best practices outlined here to protect user data, maintain system integrity, and comply with organizational security policies.

### Related Information

[Configure Identity Provider \[page 49\]](#)

Administrators can configure identity provider to enable Single-Sign-On (SSO) for the system.

# 12 Security Recommendations

Use the information in this table to secure the configuration and operation of services in your landscape.

## Note

Not all services are listed yet. We've started with core security services and are extending this list service by service.

If your service is missing, see also the *Security* section of the service documentation on the [SAP Help Portal](#).

## Security Recommendations

| Service                        | Priority | Secure Operations Map             | Topic                 | Default Setting or Behavior                                               | Recommendation                                                                                                                         | More Information                                                                                                                                                                                                                  | Last updated | Index        |
|--------------------------------|----------|-----------------------------------|-----------------------|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|--------------|
| SAP Sales and Service Cloud v2 | Advanced | Authentication and Single Sign on | Strong authentication | Default authentication method basic authentication with Username/Password | Enable SAML 2.0 Assertion for Front-End Single Sign-On and disable the default username/password authentication method (CNS-IAS-0002). | <a href="https://help.sap.com/docs/CX_NG_SV/2c87cece32844c91836e535aef8f9642/8cf5cd5ba27e408aa851196d1cf90d66.html">https://help.sap.com/docs/CX_NG_SV/2c87cece32844c91836e535aef8f9642/8cf5cd5ba27e408aa851196d1cf90d66.html</a> | 03.2023      | CNS-IAS-0001 |
| SAP Sales and Service Cloud v2 | Advanced | Authentication and Single Sign on | Strong authentication | Password access is not disabled by default                                | Disable password access for all users working with SSO                                                                                 | <a href="https://help.sap.com/docs/CX_NG_SV/2c87cece32844c91836e535aef8f9642/8cf5cd5ba27e408aa851196d1cf90d66.html">https://help.sap.com/docs/CX_NG_SV/2c87cece32844c91836e535aef8f9642/8cf5cd5ba27e408aa851196d1cf90d66.html</a> | 03.2023      | CNS-IAS-0002 |

| Service                        | Priority | Secure Operations Map             | Topic                 | Default Setting or Behavior                                                                                                  | Recommendation                                                                                                                                                                      | More Information                                                                                                                                                                                                                    | Last updated | Index        |
|--------------------------------|----------|-----------------------------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|--------------|
| SAP Sales and Service Cloud v2 | Advanced | Authentication and Single Sign on | Strong authentication | Only sample password policies are configured to showcase the options, but might not be not strong enough for productive use. | If password authentication is used (which is not recommended, refer to CNS-0001) a separate password security policy needs to be configured for administrators with stricter rules. | <a href="https://help.sap.com/docs/CX_NG_SVC/2c87cece32844c91836e535aef8f9642/8cf5cd5ba27e408aa851196d1cf90d66.html">https://help.sap.com/docs/CX_NG_SVC/2c87cece32844c91836e535aef8f9642/8cf5cd5ba27e408aa851196d1cf90d66.html</a> | 03.2023      | CNS-IAS-0003 |
| SAP Sales and Service Cloud v2 | Critical | Roles and Authorizations          | Access Control        | C4C does not ship default roles. Initial user have to create admins and all the other roles                                  | Configure a reasonable minimal privileges concept for different roles and users. Provide write access only to users who need it to operate.                                         | <a href="https://help.sap.com/docs/CX_NG_SVC/2c87cece32844c91836e535aef8f9642/fe751ce9767f49ce85954bcd0a7a3f66.html">https://help.sap.com/docs/CX_NG_SVC/2c87cece32844c91836e535aef8f9642/fe751ce9767f49ce85954bcd0a7a3f66.html</a> | 03.2023      | CNS-AUT-0001 |

| Service                               | Priority    | Secure Operations Map             | Topic                      | Default Setting or Behavior                                                                                                         | Recommendation                                                                                                                                                                   | More Information                                                                                                                                                                                                                                                                                              | Last updated | Index        |
|---------------------------------------|-------------|-----------------------------------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|--------------|
| SAP Sales and Service Cloud v2        | Advanced    | Authentication and Single Sign on | Strong authentication      | Inactive users will be logged off of the system after 1 hour                                                                        | Review the time after which the users will be log out and configure to your preference. More privileged users should have shorter timeout settings (can be configured per role). | <a href="https://help.sap.com/docs/CX_NG_SVC/56436b4e8fa84dc8b4408c7795a012c4/7b45c2825af34fee9c8847a5dd055645.html#loio59d5cf505a35433ebd00db1d8d569f36">https://help.sap.com/docs/CX_NG_SVC/56436b4e8fa84dc8b4408c7795a012c4/7b45c2825af34fee9c8847a5dd055645.html#loio59d5cf505a35433ebd00db1d8d569f36</a> | 03.2023      | C4C-AUT-0001 |
| SAP Sales and Service Cloud v2        | Recommended | System hardening                  | MIME types for Attachments | No MIME types are enabled by default.                                                                                               | Review the list of allowed MIME types and only add what you really need.                                                                                                         | <a href="https://help.sap.com/docs/CX_NG_SVC/56436b4e8fa84dc8b4408c7795a012c4/2fb39744d74a4a33a1cb8c81bb85aef9.html">https://help.sap.com/docs/CX_NG_SVC/56436b4e8fa84dc8b4408c7795a012c4/2fb39744d74a4a33a1cb8c81bb85aef9.html</a>                                                                           | 03.2023      | CNS-IAS-0004 |
| SAP Sales and Service Cloud Mobile v2 | Advanced    | Data Protection and Privacy       | Personal data removal      | Personal data and transactions can automatically be removed from the system. By default, none of the removal procedures are active. | Adjust the retention and removal procedures to your business needs.                                                                                                              |                                                                                                                                                                                                                                                                                                               | 03.2023      | CNS-DPP-0001 |

| Service                              | Priority         | Secure Op-<br>erations<br>Map       | Topic               | Default<br>Setting or<br>Behavior                                                                                                                                                                           | Recom-<br>mendation                                                                                                                                                                                                                                                                                                      | More Infor-<br>mation | Last up-<br>dated | Index            |
|--------------------------------------|------------------|-------------------------------------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-------------------|------------------|
| SAP Sales<br>and Service<br>Cloud v2 | Advanced         | Data Pro-<br>tection and<br>Privacy | Data reten-<br>tion | Logs for de-<br>personal-<br>ization and<br>disclosure<br>are kept for<br>6 months<br>by default.                                                                                                           | It is recom-<br>mended to<br>review data<br>retention<br>and define<br>other rules<br>if needed.                                                                                                                                                                                                                         |                       | 03.2023           | CNS-<br>DPP-0002 |
| SAP Sales<br>and Service<br>Cloud v2 | Recom-<br>mended | Security<br>monitoring              | Monitoring          | Monitoring<br>and alerting<br>is a shared<br>responsibil-<br>ity in which<br>SAP fo-<br>cuses on in-<br>frastructure<br>level events<br>and cus-<br>tomers fo-<br>cus on the<br>application<br>level events | Security-<br>relevant<br>events to be<br>reviewed<br>regularly:<br>Settings -<br>Data Ad-<br>ministration<br>- Audit Logs<br><br>Logged<br>events:<br><br>Failed and<br>successful<br>login<br><br>User crea-<br>tion and up-<br>date<br><br>Password<br>update<br><br>Authoriza-<br>tion check<br>success<br>and failed |                       | 03.2023           | CNS-<br>MON-0001 |



# Important Disclaimers and Legal Information

## Hyperlinks

Some links are classified by an icon and/or a mouseover text. These links provide additional information.

About the icons:

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